



Decoding comics: a systematic literature review on recognition, segmentation, and classification techniques with emphasis on computer vision and non-computer vision

Rishu¹ · Vinay Kukreja² 

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Abstract

The increasing popularity of digital comics in recent years has drawn substantial interest in comic recognition (CR). CR is a process that includes the automated recognition and extraction of comic panels, speech balloons, and text from comic images and covers a wide range of applications within the digital media sector. This study aims to thoroughly review the existing literature on CR methods and techniques to present all significant findings in recognition techniques, datasets, and comparative evaluations of recognition models. Systematic Literature Review (SLR) is conducted using a search strategy that includes various databases, and 60 studies from 2011–2024 are selected. Most of the studies examined for this study use computer vision (CV) techniques for CR, which includes 65% of studies. The remaining 35% of studies have used non-CV techniques. The studies employed segmentation techniques (40%), machine learning (28%), and deep learning (32%). Manga109 and eBDtheque are the most popular public datasets that 78% of the selected studies used. The remaining 22% of the studies built their datasets using existing datasets. The study summarises the findings of the CR research and emphasizes the need to establish uniform standards for accuracy and datasets. It highlights the necessity of investigating and creating hybrid approaches for effective CR operations.

Keywords Deep Learning · Machine Learning · Comic recognition (CR) · Computer Vision (CV) · Non-Computer Vision (Non-CV)

✉ Vinay Kukreja
onlyvinaykukreja@gmail.com

Rishu
rishutaneja066@gmail.com

¹ Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India

² Centre for Research Impact & Outcome, Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India

1 Introduction

The world is familiar with graphic novels like comic books and manga. Comics are now increasingly being read on smart phones and tablets rather than on paper due to the shift to the digital age [1]. Several types of comics-related studies have been put out recently, and they may impact how comics are produced, disseminated, and viewed in the future [2]. Comics were formerly regarded as "children's literature" or "sub-literary" since they combine text and visuals. Still, they have recently attracted much attention due to their sophisticated use of graphics to express concepts and aesthetics. Comics are visual writings that profoundly impact their readers, both adults and children and assist in drawing in a sizable audience [3]. Utilizing the most recent technology makes it possible to effectively make digital copies of comic books available for free to every reader, anywhere, at any time [4]. Thus, there is a need to work on comic books so that they may be given to their user in digital media in a user-friendly and inventive way. Independent research on comics has been done in a variety of fields, including multimedia, document image analysis (DIA), human–computer interaction (HCI), and others [5]. It is essential to be able to extract and identify parts of comic books to ensure that digital comics can be accessed in a manner that is accurate and user-friendly across every platform [6]. As stated by [7], more research into the connections between these components might likely be conducted to assist a computer in comprehending comic books in their digital form. Using this technology, the user can get more precise information from the picture corpus. Several components, including panels, balloons, text, and comic characters and their relationships, make up comic book visuals (e.g., read before, said by, thought by, addressed to). For nearly a decade, the image analysis community has been exploring the extraction of comic book elements [8]. The approaches used range from low-level analysis, including text recognition, to high-level analysis, including style recognition. One of the most researched aspects of comic book picture analysis is text detection and identification. The authors suggested several strategies based on image processing techniques [9]. In a recent study, the authors presented methods for text recognition that do not need text letter segmentation. These methods are based on deep learning models [10]. One of the most complicated challenges is identifying comic characters (protagonists) since comic book writers have complete creative freedom when creating their characters [11]. As a result, the characters' appearances might vary significantly across comic books and even inside individual comic books. Several approaches based on deep neural networks or manually constructed feature-related strategies have been put forward for the recognition of comic book characters or faces [12]. One essential component of comics is the speech balloon, which may take on various forms. The balloon extraction methods now in use are based on traditional image processing approaches like contours and area recognition [13]. The technique described may likewise link speech balloons to humorous characters.

Research on panel extraction has been ongoing for a while. Accurate panel extraction has become more critical lately because of the advancements in the size and quality of screens on mobile devices, such as tablets and smartphones [14]. Techniques in this field include region growth, morphological analysis, linked component labelling, and white line cutting [6, 15]. Recently, novel techniques have been developed based on area of interest identification, recursive binary splitting, and watershed line segmentation employing the canny operator and polygon detection. By representing the identified panels using a quadrilateral form rather than the bounding box, the work in has suggested novel methods that can handle irregular comic panels [16]. Every approach now in use, whether it is based on traditional techniques or deep learning models, handles each comic element independently.

Since there is barely an algorithm that can extract every element from comic book pictures simultaneously due to their unique components, this method has been adopted [17].

1.1 Highlights of the study

- Offering an understanding of the growing importance of CR in the field of digital comic content.
- The study thoroughly explores the broad range of CR techniques, including computer vision and non-computer vision techniques.
- Offering an in-depth understanding of the wide range of CR techniques, including segmentation, machine learning, and deep learning techniques.
- Investigating the scope of public and private databases that are available for CR.

1.2 Motivation for the work

- First, there is a growing amount of scholarly study on comics, a medium that is becoming increasingly popular and important to society. A thorough overview of the majority of information regarding comics may be provided by this systematic literature review (SLR), which can also assist in synthesizing this study.
- Researchers, educators, and professionals who use comics in their work or are interested in examining how comics have affected many facets of society may find this to be helpful. Additionally, a systematic review might point out areas for future study and assist in filling in research gaps, thereby advancing the subject of comic studies.

The structure of the rest of the paper is as follows: Section 2 presents the history, applications, language, recognition models, and theories for the analysis of comics. The research methodology utilized to examine the analysis of comic images for the current study is described in Section 3, along with the analytical framework that has been chosen. The findings and discussions of a systematic review are presented in Section 4. The theoretical and practical implications of comic recognition have been discussed in Section 6. The conclusion and future scope for the comic recognition has been discussed in Section 6.

2 Background

In the academic discussion of comics, several aspects are meticulously examined, creating an extensive network, as shown in Fig. 1. They are reviewing comics, a crucial part of this research. The authors carefully analyze this one-of-a-kind medium's narrative and creative aspects, trying to make sense of the many meanings conveyed by the combination of images and text in comics. Critical to this academic study is the structural examination of comics [18]. Researchers explore the complexities of panel arrangement, timing, and sequencing to decipher the visual language that supports narrative. In addition to illuminating the compositional decisions made by comic book artists, this analytical effort expands our comprehension of how these choices affect the story as a whole. Comics are helpful as a powerful medium for communication in many different areas, going beyond their entertainment value [19]. Comics have many practical applications, and researchers investigate their uses in fields as varied as healthcare, activism, and education [20]. This inquiry



Fig. 1 Facets of comic

shows the adaptability of comics as a medium that goes beyond conventional limits and connects with a wide range of viewers. Researchers in the field of comics use recognition models to build frameworks for understanding and appreciating the distinctive aspects of this visual language. By breaking down the medium into its constituent parts, these models help researchers better grasp the social and cultural effects of the medium as a whole [21]. Researchers are captivated by the language of comics, a unique combination of verbal and visual components. The authors closely examine the medium's linguistic intricacies to see how its artists use a rich vocabulary of words and visuals to tell intricate stories [9]. Studying this distinct language deepens our understanding of the expressive power of comics. All the comic facets have been discussed in more detail in the sub-sections below.

2.1 Evaluation of Comic

A comic can also be used as a countable noun, in which case the specific example of the medium, such as a comic book or comic page, is referred to. Along with television, radio, etc., comics are a means of disseminating information [22]. There are essentially no guidelines for making comics, as for any art. Anything, however, they choose to draw is up to the researchers [2]. Even though comics have been present for centuries, the contemporary comic book as we know it today started to develop in the early twentieth century. This is a synopsis of how comics have changed over time, as shown in Fig. 2.

- Early comics in newspapers (the 1890s–1930s): The first comics were usually single- or multi-panel strips published in newspapers from the 1890s through the 1930s. These comics often included straightforward line drawings and comedy [23].
- Golden Era of Comics (the 1930s–1950s): In the 1930s, DC Comics and Marvel Comics first appeared, introducing Superman, Batman, and Captain America. With the addition of new subgenres like superheroes, horror, and science fiction, comics grew in popularity and diversity [24].

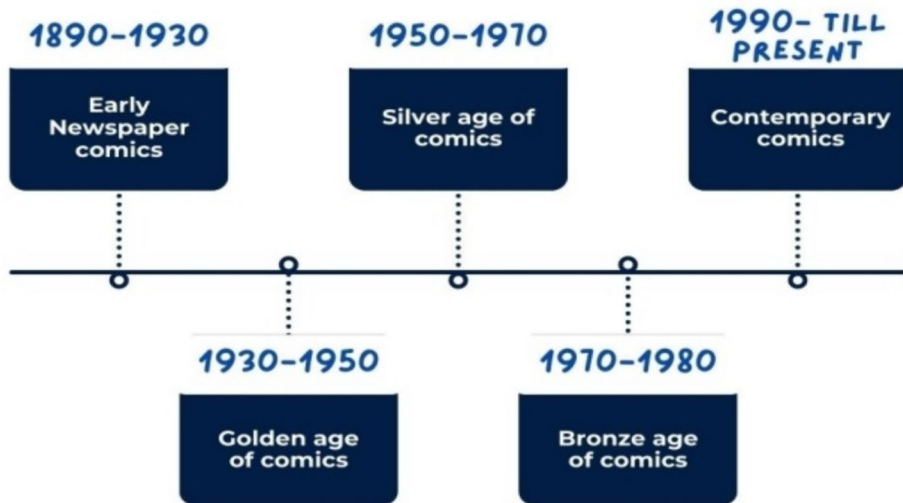


Fig. 2 Evolution of comics

- Comics' Silver Age (1950s–1970s): New superhero characters, including Spider-Man, the X-Men, and the Fantastic Four, were launched during this period, which witnessed the resurrection of superhero comics. Along with developing increasingly intricate plots and artwork at this time, the Comics Code Authority was also established to control the content [25].
- Bronze Age of comics (the 1970s–1980s): Plots and characters tended to be more socially conscious, and diversity and representation were prioritised. In terms of plot and aesthetics, comics also evolved to become more experimental [26].
- Contemporary comics (since the 1990s till the present): Graphic novels and webcomics are two examples of new comic book forms and styles that have emerged in the modern era. Moreover, indie publishers and creator-owned comics have received more attention. The medium has also been significantly impacted by digital technology, as comics are now accessible in digital formats and on interactive platforms [27].

Generally, there have been significant changes throughout the history of comics, with new forms, genres, and styles appearing over time. Despite these developments, comics have continued to captivate readers' imaginations worldwide, and the medium has remained a popular and significant form of storytelling. Let's discuss the structure of the comic image in more detail with the elements present in the comic image and the use of these elements for representing the narrative structure of the comic image in the below section.

2.2 Structure of Comics

A complicated kind of document is comics. Most of the time, they are released as printed periodicals or books or integrated into other media, including newspapers. Short comic strips are typically seen in comic books, which may convey complicated storylines across several pages[28]. Each page in a comic book comprises one or more separate pictures placed in a layout and

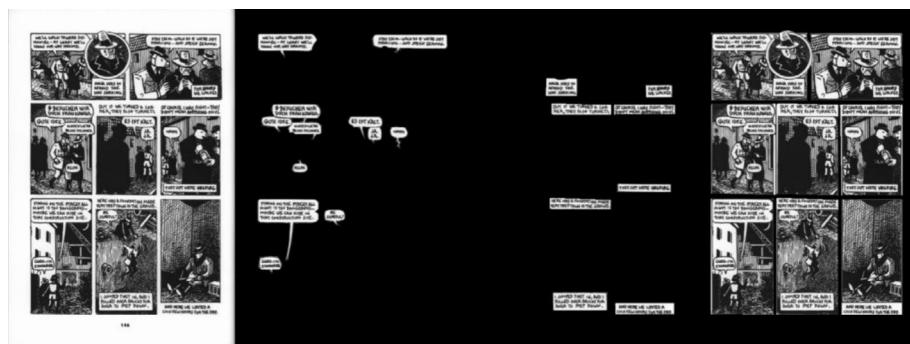


Fig. 3 Ground truth annotations for captions, speech bubbles, and panels on a comic page (from left to right) [35]

frequently enclosed by created visual borders, as shown in Fig. 3. These so-called panels serve as a crucial structural and linguistic unit in comic books. They often include textual elements like speech balloons and narrative captions, which indicate narrative text and a specific character's speech or thoughts, respectively, as well as line drawings of characters, objects, and backgrounds that are depicted in a visual scene [29]. Other components that visualize sound may be added, such as motion lines, diegetic writing, and onomatopoeia. Then, some components represent significance, such as gears turning above a character's head to show that they are thinking. Finally, in the comic book world, some symbols convey meaning. For example, a character's head may have gears moving above it to represent thought. According to [30], these are lexical items in a visual language that may be combined with other words. The term "visual morphemes" refers to elements such as "pictorial runes" [31] and culturally particular elements such as the "Manpu" (small symbols used in Japanese manga that are skilled at expressing the feelings and senses of the characters). Artists are empowered to express themselves in many ways when creating the composition and spatial arrangement of panels on a page, which make up a page layout. These designs range widely from simple grid layouts to more complex ones with staggering panels slightly shifted horizontally or vertically about other elements like background images, insets, bleeding, and panels separated by imaginary contours [32]. Page layout complexity has changed with time in line with other aspects of the narrative artistry seen in comics. However, one crucial element is still these designs' navigability, which guarantees that the panels are understood in the correct order, a factor that is partly impacted by widespread cultural norms [33]. The sequential organization that characterizes comics is a significant feature of their work, leading to the term "sequential art" used in comics [34]. Known as the gutter, the space between panels requires readers to do a mental act of inference to go smoothly from one panel to the next.

One of the first corpus studies in this field is provided by [36], who is well known for his wise examination of comic art. He presents a taxonomy that defines panel transitions. Computational analysis of panel transitions is a difficult task since the inference complexity needed depends heavily on the semantic richness of the panels. More than that, comic book readers must build a mental model beyond the fundamental panel-to-panel changes [37]. A thorough framework clarifying the structures inherent in visual narratives is introduced [9]. In this paradigm, a unique mode of expression graphics is used to convey conceptual information, capturing meaning using combinatorial systems comparable to grammar. These structures function at the narrative level at both the unit and sequence levels [37]. Each panel uses a visual language at the unit level, and the layout's panel arrangement coordinates the reader's step-by-step

experience. Visual tales need the viewer to develop scenario and event models since, conceptually, they incorporate meanings [2]. The method is building story units using standardized visual morphological elements similar to morphological elements in the language. Once the fundamental structural aspects of comics have been fully understood, it is necessary to explore how comic pictures are used in various contexts. Comics are more than simply a means for artistic expression; they are a strong medium that can be used in a wide variety of contexts because of the mix of dynamic visual features and narrative components that give them a deeper meaning.

2.3 Applications of Comic

As a result of the fact that comics are a one-of-a-kind blend of artistic flair and narrative intricate details, they have the potential to be used in a broad range of scenarios. Because of the unique combination of captivating visuals and great storytelling abilities, comics can go above the restrictions that have traditionally been associated with them to establish themselves as a powerful medium [38]. Comic books are equipped with incredible adaptability, enabling them to extend their influence beyond the restrictions often associated with their medium [39]. This is because the narrative depth and beautiful images perfectly harmonise. Comic books have arisen as a very effective medium for disseminating information, engaging audiences, and encouraging creative expression as a result of [40]. The intrinsic synergy between visual and narrative aspects positions comics as dynamic and flexible vehicles of communication and creative expression across various industries, including but not limited to education, marketing, and entertainment, with their advantages shown in Table 1.

After examining the many uses of comics in various fields, it is necessary to understand the techniques used to extract multiple elements from comic images. To improve the study of comic book visuals and provide a deeper look at the narrative they tell, this in-depth examination is of the highest significance. In extracting information from humorous drawings, three fields, artificial intelligence, image processing, and computer vision, are intimately intertwined. Regarding the realm of comic pictures, the next part will explain the basic techniques utilized to extract different parameters.

2.4 Comic Recognition models

Using a wide range of methodologies and algorithms, the phrase "comic image recognition models" describes computational systems designed to detect and interpret visual elements included inside comic pictures [46]. These models, which are crucial components in this method of automating the research and analysis of comic book material, make it feasible for such applications as content retrieval, genre classification, and character identification to be carried out [47]. Building these models necessitates using computer vision and machine learning techniques to recognize and categorize the many components present in comic panels. Table 2 below lists the many comic book recognition models various authors used in their research.

The usage of comic book recognition models is growing in academic and industrial settings. They may be utilized to automate the process of comic book analysis, minimizing the time and effort needed to glean insightful information from the content. Table 2 offers an extensive overview of the many aspects of the computational analysis of humorous images. Advanced methods like CNN and OCR are used in these procedures, from object identification and panel detection to sentiment analysis and cultural context recognition. These

Table 1 Use of Comics

Application Field/Reference	Description	Advantages
Entertainment Industry/ [41]	Comic books are the inspiration for many film and television adaptations; this is especially true of the success of superhero series like DC and Marvel	■ Attracts over audiences with visually stimulating narrative ■ Makes use of preexisting comic book fandoms to attract people
Education / [42]	Comics serve as instructional instruments, enhancing reading abilities, engaging readers who are cautious, and efficiently communicating complex ideas across many disciplines	■ Enhances the ability to understand and interpret visual information ■ Appeals to diverse learning styles ■ Injects a captivating and pleasurable element into the process of acquiring knowledge
Marketing and advertisement / [43]	Comics are skillfully used in advertising campaigns to create memorable and lasting material that highlights product attributes and communicates brand messaging	■ Captivates the attention of the viewer by using visually appealing storylines ■ The power of narrative plays a significant role in increasing brand retention ■ Provides customers with the opportunity to form emotional ties
Health Communication / [44]	Comics provide a significant contribution to the field of health communication by reducing complex medical topics in order to improve awareness across a variety of audiences	■ Increases health literacy by making use of various visual aids throughout the process ■ A more sophisticated understanding of medical information is made more accessible ■ Improves the patient's ability to participate and adhere to the medical advice given
Journalism / [45]	Comics journalism, also known as graphic journalism, is a unique medium for providing full news coverage through the seamless integration of pictures and text	■ Creates an engaging experience for readers by using a novel way of storytelling ■ The provision of a visual story complements the use of standard journalistic approaches ■ It enhances the reader's ability to retain information and comprehend it
Storytelling platforms / [31]	The reach of comics is expanded by the use of digital platforms and webcomics, which provide authors with innovative pathways to spread stories online. These avenues include all of the different social media platforms	■ Through the use of internet platforms, it supports worldwide reach and accessibility ■ In the comic book business, it encourages the development of different and autonomous voices ■ A direct link is established with the communities and readers

Table 2 Recognition techniques

Reference	Aspect of Recognition	Description	Technique	Application
[48–50]	Object identification	Classification and differentiation of elements in comic panel representations, including backgrounds, objects, and characters	Convolutional Neural Network (CNN), You only look once (YOLO)	Character tracking, object-based narrative analysis, style analysis, and scene interpretation
[8, 14, 16, 33]	Detection of Panel	They identify and separate individual panels from comic pages so that sequential structures can be analyzed methodically	Connected component labelling (CCL) algorithm, contour detection algorithms, U-net, CNN	Improved understanding of layout, navigation support, and sequential analysis
[11, 31, 51]	Comic character recognition	Used for the recognition of specific characters in comic books	CNN, Thresholding, Region-based Convolutional Neural Network (R-CNN)	assistance with sentiment analysis, visualize studies, and character tracking
[52–54]	Extraction of text	Take the text out of comic panels so that onomatopoeic phrases, dialogue, and captions are more easily analyzed	Optical Character Recognition (OCR), CNN	Dialogue analysis, narrative comprehension, and language translation
[55–57]	Speech balloon	To accurately find the speech balloons, which are usually speech bubbles containing sound effects, dialogue, or thoughts from the characters	Morphological operations, CCL, Clustering, Watershed segmentation	Dialogue understanding, narrative analysis, content summarization, translation services
[42, 58, 59]	Analysis of page layout	It is understanding the panel and gutter positioning, arrangement, and other visual components of a comic page	Clustering techniques, Grid detection algorithms, Edge-based segmentation	Enhanced understanding of pages, examination of visual storylines, and assistance with narrating
[3, 9, 45]	Sentiment and Emotion analysis	A story's emotional tone is evaluated by identifying body language, facial expressions, and contextual clues	CNN, Electroencephalogram (EEG), E4 wristband	Emotional tone assessment, character emotion monitoring, and narrative influence
[47, 60, 61]	Genre classification	Visual clues, colour schemes, and creative approaches distinguish comic book genres	CNN, ResNet50	Recommendations for genre-specific content and focused audience interaction
[59, 62]	Recognition of cultural context	Recognizing cultural components in comics, such as visual themes, symbols, and artistic traits exclusive to specific cultural settings	Pattern recognition, CNN	blending context-aware storytelling, cross-cultural study, and cultural storytelling

sophisticated methods support more complex tasks, including character tracking, narrative understanding, emotion analysis, genre-specific material suggestions, and identifying cultural quirks in comics. Combining these cutting-edge approaches highlights how complex comic image analysis is, improving our ability to decipher the nuances of visual stories and derive meaningful conclusions from the diversity of components present in comics.

After explaining techniques used in comic image recognition models, the next part will investigate the several languages inherent in the comic book medium. This talk seeks to highlight the unique language components and visual lexicons that enrich the complex structure of the comic book medium. The writers aim to offer readers a thorough understanding of the complex linguistic diversity deeply integrated into the visual storytelling within the comics industry by carefully examining the various cultural, artistic, and narrative nuances that emerge within different comic languages.

2.5 Languages of Comic

Since comics are a multilingual narrative medium, they highlight this storytelling style's universal appeal and cross-cultural variety [38]. Comic books are widely available, which is evidence of their universal appeal. Table 3 illustrates the linguistic diversity of the medium by presenting the most significant languages in which comics are created both widely and passionately. It also thoroughly explains the main languages associated with various humorous traditions, locations, and distinctive characteristics. To engage a large and culturally varied audience, it visualises the global diversity found in the comic book medium, demonstrating its accessibility across several languages.

3 Method to conduct SLR

To conduct the systematic literature review (SLR) to analyze the comic images, which include speech balloons, panels, narrative text, gutters, comic characters, etc., the guidelines of Kitchenman [63] have been followed which is used to provide the overview and detailed description of the protocol for conducting SLR. According to the protocol, the authors followed various steps to execute the review protocol efficiently, as shown in Fig. 4. The steps for implementing the protocols are as follows:-

- Formulation of research questions
- Designing a search system
- Selection of studies
- Quality assessment of studies
- Extraction of data
- Synthesis of data
- Threats to validity

The review methodology states that the first step outlines the research questions connected to the detection and extraction of comic images. The search approach that intends to assess the papers that will aid in elucidating each research issue is described in the second phase of SLR. The search string and literature sources used to define the iterative and unbiased search approaches are part of this step. Based on research questions for that relevant study, the third step has been chosen among all searched literature studies. In essence, the

authors have determined requirements for inclusion and exclusion criteria to determine which research study should be included and which should be eliminated during this phase. The strength and character of each investigation are calculated using criteria for qualitative evaluation. In the data extraction phase, the authors collect all the relevant information based on research questions. The last and final stage describes the methodology and synthesis of the data collected depending on the research questions. Developing a review methodology is a crucial step in SLR since it lowers the possibility of research bias. The authors provide a thorough explanation of each phase of our SLR in the subsections that follow.

3.1 Research questions

The objective of this SLR is to analyze and evaluate the significant data obtained from several types of research that used various techniques to detect and extract different parameters (e.g. panels, speech balloons, text, character, tail) from comic images. The authors have developed and examined the following research questions, as shown in Table 4. A guiding light that orchestrates the intricate orchestration of the whole review process is the research question. This crucial element plays a vital function by precisely defining the review's parameters and scope, giving the analysis a defined direction and purpose. A well-crafted, specifically focused search strategy that skillfully selects relevant keywords and databases necessary for an exhaustive study mostly depends on the research issue. The research question also plays a crucial role in helping to define the strict inclusion and exclusion criteria, which guarantees that the chosen studies will align with the review's stated goals. As studies are carefully screened and selected, the research question is a critical reference point that judges each one's eligibility. Reviewers are guided to material closely related to the primary investigation by the research question throughout the data extraction phase. Results are presented with a natural framework and coherence via the synthesis and analysis of gathered data that follow, which are closely linked to the study objectives.

3.2 Search strategy

In systematic literature reviews, search strategies refer to the organized and well-planned process of finding relevant research within the predetermined review scope. Ensuring that all suitable material that directly answers the review's goals or research question is seen is the aim of the search method. This is the next phase of SLR that describes search strings, database selection, and the search process for the corpus collection. The detailed description of each sub-phase is as follows:-

3.2.1 Search string

When creating a query for a search in a database or search engine, a search string is a collection of keywords and operators. It is an essential feature of any research strategy's search engine optimization, especially for SLR or other academic enquiries. Choosing pertinent phrases, using Boolean operators, and considering synonyms or variants relevant to the research issue are all essential steps in creating a successful search string. The creation of a search string depends on various steps, which are mentioned below:

Table 3 Comic languages

Language	Comic Styles	Publishing Country/Region	Characteristics
English	Renowned for webcomics, graphic novels, and superhero comics	Most often in Canada, the United States, and the UK	Maintains its dominance as the primary language for popular superhero comics, with wide-ranging international resonance
Chinese	Renowned for translating manga and webcomics	Hong Kong, China and Taiwan	It is growing in popularity, providing a wide range of translated books and comics
Italian	Recognized for Fumetti's comic art	Italy	Upholds the strong fumetti heritage while highlighting the subtleties of pictorial narrative
Japanese	Well-known for its Manga designs	Mostly from Japan	Have versions in many other languages and an extensive worldwide following
French	Particularly to the customs of Bandes Dessinées	Belgium and France	Throughout Europe, French comics are widely available and represent a variety of genres
Korean	Notable for its Manhwa storytelling style	South Korea	They are gaining increasing recognition on a global scale, especially with English translations
Spanish	They are known for their comic strips and graphic novels	Latin America and Spain	Features a vibrant humour culture enhanced by a rich historical legacy in Spanish-speaking areas
Portuguese	Comic strips and Graphic novels	Brazil, Portugal	Portuguese-language comics enrich the cultural story in these locations
German	They are known for their comic books and graphic novels	Switzerland, Australia and Germany	We are seeing an increasing number of comics geared toward German-speaking readers
Russian	Renowned for webcomics and graphic novels	Russia	It is generating more interest in a wide variety of genres of Russian-language comics
Arabic	Well-known for webcomics and graphic novels	North Africa and the Middle East	It becomes increasingly popular as the amount of visual narrative being produced increases
Hindi and other Indian Languages	Recognized for webcomics, comic strips, and graphic novels	India	Showcasing the Indian subcontinent's rich comic culture, this is created in various languages

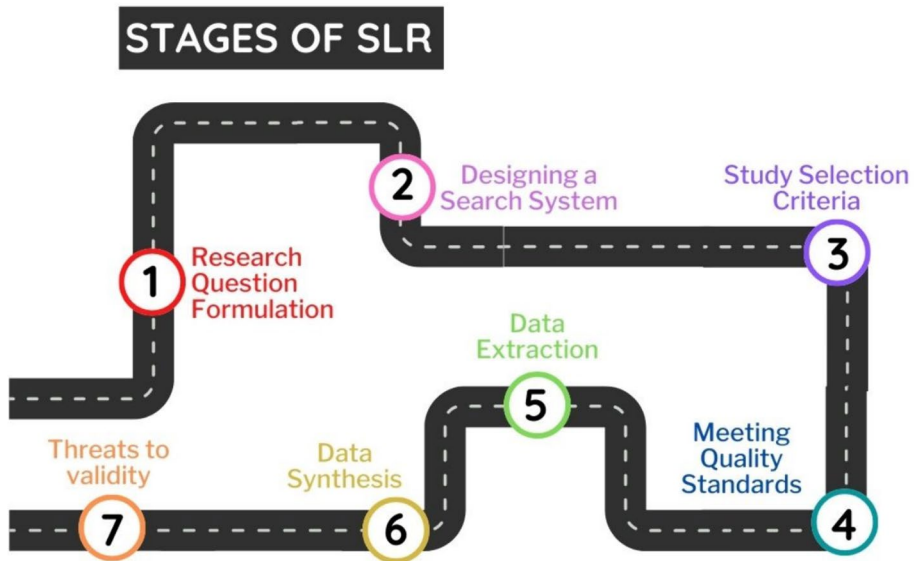


Fig. 4 Stages of SLR

- To identify the significant string that is extracted from research questions.
- To identify the equivalent for each significant string.
- Examine the titles of books or articles that are linked.
- OR operator is used as equivalent words.
- The Boolean AND operator combines the significant string with equivalent words.

The authors created a more efficient search string by combining the Boolean operations “OR” and “AND” with includes all synonyms while considering comics SLR. The authors have picked up comics or comics and combined them with the search term. In addition, the authors have combined extraction, detection, segmentation, recognition, and analysis. The final search string is:

((COMIC OR COMICS) AND (EXTRACTION OR DETECTION OR RECOGNITION OR SEGMENTATION OR ANALYSIS))

The secondary search will now be conducted by examining the references of the articles chosen through the search string after the search string has been applied to the specified databases.

3.2.2 Database selection

When doing SLR, choosing a database is essential in the research process. It entails selecting databases carefully to be most relevant to the topic or field of study. To improve the chances of discovering highly relevant articles, a suitable selection of databases must be picked before beginning the search process. Based on the databases’ content coverage, applicability to the research question, and tendency to hold research papers, the authors have chosen specific databases for their work, listed below. To ensure thorough coverage and customize the search to the distinct features of each desired database, it is essential to

Table 4 Research Questions

RQ#	Question	Motivation
RQ1	What are the different parameters of comic images used to detect the information from comic images?	Identify different types of parameters of comic images
RQ2	What are the different types of speech balloons used in comic images?	It describes the different types of balloons to provide meaningful information from comic images
RQ3	What are the different computer vision techniques used for comic analysis?	To explore different computer vision techniques that are available for comic analysis
RQ4	What are the different non-computer vision techniques used for comic analysis?	It helps to describe the various non-computer vision techniques available for classifying comic images
RQ5	What are the different computational approaches to comic analysis?	It helps to identify different approaches available for comic analysis
RQ6	What are the different types of datasets available?	It describes the different types of datasets available publicly for comic research
RQ7	What are the different genres of comic images?	It helps to explore different tools which are available for the extraction of comic images
RQ8	How can the emotion of a character be recognized through comic images?	It describes different parameters that help recognize the emotional state of a comic character through speech and video analysis
RQ9	Which Journal/Conference publishes the most comic detection and extraction research	To identify different journals and conferences available for publishing research work
RQ10	What are the different accuracy measurements used for a comic analysis?	It helps to describe the various accuracy measurements

cast a broad net. To identify the relevant articles, the authors have selected the following databases:

- IEEE Xplore (ieeexplore.ieee.org).
- ACM Digital Library (www.acm.org/dl).
- Springer (www.springerlink.com).
- Google Scholar (<https://scholar.google.com/>).
- Scopus (www.scopus.com).

3.2.3 Search process

To find appropriate information or a particular research article, a search string has been used mainly on the database (i.e. IEEE Xplore, Google Scholar, ACM, SCOPUS, and Springer), and a secondary search has also been conducted. The research works that will be finalized for SLR were selected based on the abstract, title, and keywords. A total of 1,510 studies were identified during the identification phase, as shown in Fig. 5. After these studies underwent screening (i.e., duplicate articles were removed), the authors were left with 890 research studies. The next step consisted of the eligibility phase, where 250 research studies remained after the authors followed the inclusion and exclusion criteria. In the following phase, the authors conducted a quality assessment, and the remaining 60 research studies were further selected and included in SLR research studies.

3.3 Selection of research studies

After conducting the search process successfully and choosing the research articles based on their titles and abstracts to obtain more credible and relevant literature, the authors have employed two primary steps for further filtration of the research studies. The two preliminary steps include:

In the first step, the authors chose the inclusion/ exclusion criteria for the filtration of the research studies, i.e., which research study will be included or excluded from the SLR. After that, in the next step, the authors performed a quality assessment checklist to check the quality of the research articles.

Both the steps chosen by the authors with their working are well explained in the following sub-sections below:

3.3.1 Inclusion and exclusion criteria

For the selection of research studies, the first step that the authors have chosen is the inclusion/exclusion criteria. These criteria serve as discriminating filters, methodically directing the inclusion or removal of research studies according to predefined characteristics, guaranteeing methodological consistency and conformity with the study goals. The inclusion criteria specify the necessary qualities that research should have to be considered for inclusion in the review. These qualities might include things like the study design, the kind of publication, or specific demographic requirements, as shown in Table 5. On the other hand, exclusion criteria describe the characteristics or circumstances that make research unsuitable, including methodological flaws, inappropriate scope, or publishing in

non-peer-reviewed journals. Reviewers are provided with a systematic framework to filter through the accumulated literature and choose a cohort that satisfies the highest methodological rigour and relevance standards by the precise and aligned criteria with the research goals. This selective procedure guarantees that the final selection of studies in the SLR results from a careful and targeted compilation, strengthening the overall credibility and dependability of the review.

The authors chose qualitative and quantitative research articles published from 2011 to 2024. After determining the inclusion/exclusion criteria, the authors are left with 250 research studies that go further for the quality assessment checklist in the next step.

3.3.2 Quality assessment checklist

As explained by the authors [64], the selected studies' quality assessment (QA) serves as the first basis for allocating weight to the recovered quantitative data during the meta-analysis. This is because the QA measures the statistical significance of the data. One of the most critical aspects of synthesising essential data is the strategic use of quality assurance, which aligns with the inclusion/exclusion criteria set to discover relevant publications. After completing this screening procedure, the authors tested the remaining research studies through quality assessment. The authors have used the results to enhance the interpretation of the review findings and signal the robustness of the conclusions, which ultimately leads to increased objectivity in the review. The entire list of questions for the quality assurance test and the summary corresponding to each question is included in Fig. 6. The authors, in essence, made strategic use of quality assurance not only to strengthen the inclusion–exclusion procedure results but also to validate the rigorousness, trustworthiness, and relevance of the studies chosen for further examination. The QA checklist depends upon the list of questions mentioned for each section in Table 6. If a positive reply

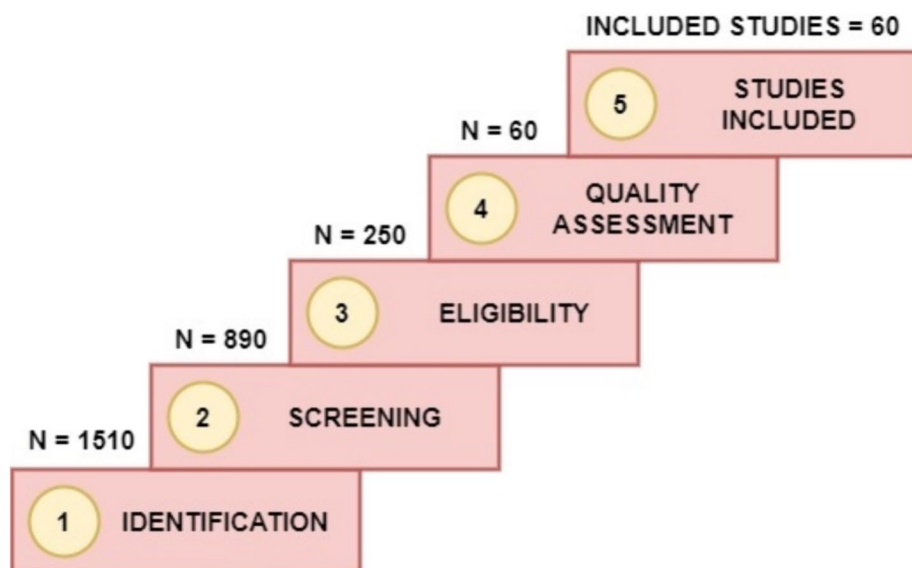


Fig. 5 Search Process

is received from Phase 1, only it can proceed to Phase 2; otherwise, the research study will be rejected. For example, if the research study effectively answered both the questions of Section 1, then only the research study will go to Section 2 for further evaluation; otherwise, the study will be rejected.

After carefully examining the quality evaluation criteria, 60 papers from 2011 to 2024 were chosen via a careful curation process. Figure 7 visually represents how the studies are distributed each year.

3.3.3 Extraction of data

This review altered selected research studies by extracting data to gain valuable information. This results in a response to the research questions used in this review. The authors discovered that some researchers used synonyms for comic analysis techniques. For instance, in various studies, the authors have used “segmentation”, some have used “extraction”, some have used “recognition”, and some authors have used “analysis”. These research studies can be regarded as subclasses of comic analysis techniques. The authors have analyzed the citations of each research study and divided them into citation brackets. As an example, there is a collection of 30 studies with citations between 0 and 20, 13 studies in the 21–40 range, seven studies in the 41–60 range, four studies in the 61–80 range, and three studies in the 81–100 and 101 and above categories. This analysis mentioned in Table 7 provides an in-depth overview of the distribution of citations for the chosen research studies and an easy-to-understand explanation of the frequency of citations in the indicated brackets.

After analysis of the citations of the selected research studies, the authors gave credibility scores to each research study on some defined criteria. For example, each study will be given a score based on the research questions to indicate its reliability. The authors have chosen a range for assigning scores, i.e., from 0 to 10, in which 0 will be considered the minimum score, and 10 will be considered the maximum score for each selected research study, as shown in Table 8. For instance, every research study obtains one score for answering each question. The authors have ten research questions in this SLR, so the selected research study can get ten marks for answering the research questions.

The purpose of giving marks to each research study is to suggest certain high-quality studies based on their total score to our readers. Table 8 displays all of the research studies along with the credibility score. According to the analysis given above in Table 8, the highly-scored studies are S4, S5, S7, S52, and S54, whereas the lowest-scored studies are S26, S28, S34, and S46, respectively. The authors have concluded that none of the selected research studies address the entire research question. The credibility scores are assigned based on the number of research questions each research study answers. Consider S26; this research study answered the two research questions in their research study, so the authors gave the marks 2 out of 10; similarly, for each research study, the marks were allocated. Let's consider S4. This study is a survey study, so it focuses on all available datasets for comic image analysis, discusses the segmentation and classification techniques, all the performance metrics, etc. So that's why the authors gave this research study a credibility score of 10. This method was followed to provide the credibility score for each research study. Further, the authors have discussed the studies that have been highly cited, i.e. (lies between 80–100), with their advantages and imitations in Table 9.

Table 5 Inclusion/ Exclusion Criteria

S. No	Inclusion Criteria	Exclusion Criteria
1	Studies that are based on comic image analysis	Studies that are not related to the humorous analysis area were excluded
2	Studies that are available in full-text access	Studies that are without full-text access
3	Quantitative and qualitative research studies from the year 2011 published till the year 2024	Extended versions of some of the research studies that first appeared in conferences before appearing in journals; these exploratory studies were excluded
4	Studies written in the English language were included	Position papers were excluded from SLR

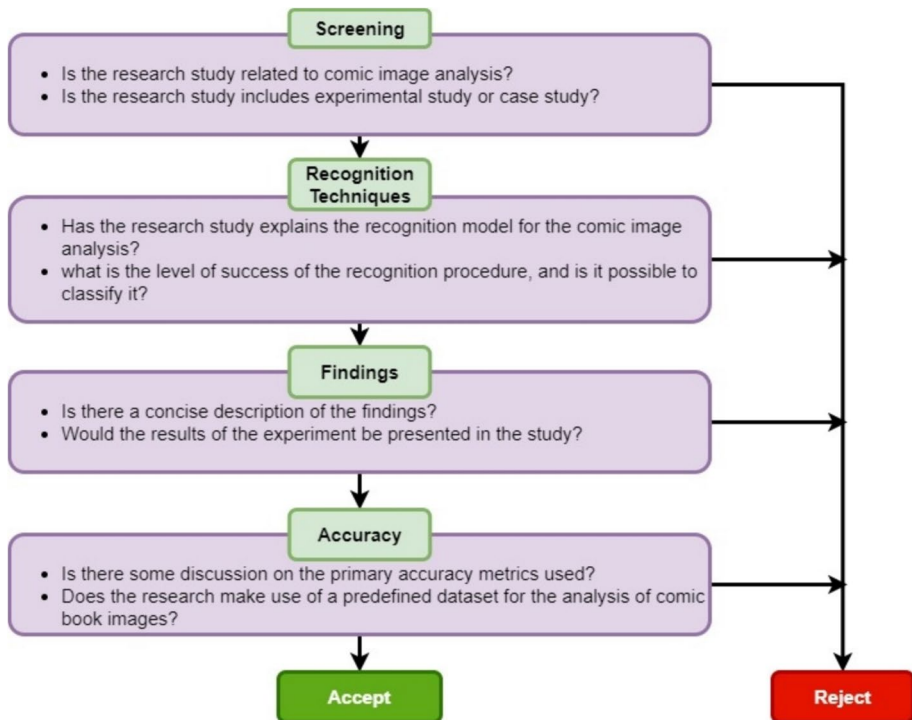


Fig. 6 Framework for Quality Assessment

3.4 Data extraction

This section gives a detailed description of the systematic data retrieval and subsequent explanation of information derived from the collected and selected research. Data extraction from the chosen research studies has been carried out with careful consideration to provide helpful information, which will then be analyzed to conclude. This data extraction and subsequent information distillation process is conducted to answer the research questions (RQs) that have been developed. Besides the initial phase, data extraction is a continuous process that requires careful monitoring and analysis. The first thing to do is to extract information from the research to determine which can answer the research question. Table 10 provides the detailed data extraction technique that shows how each study responded to specific research questions.

3.5 Synthesis of data

The primary role of this step is to find relevant information from all the selected studies that will be helpful for the research questions of this SLR. A particular amount of data might not be sufficient as evidence, but several pieces of information combined can reinforce a statement. We used two techniques to combine the resultant data:

Table 6 Quality assessment criteria

Section	Question	Evaluation
Screening	• Is the research study related to comic image analysis? • Does the research study include an experimental study or a case study?	If a response is yes, go to Sect. 2; if not, reject the research study
Recognition Techniques	• Has the research study explained the recognition model for the comic image analysis? • What is the recognition procedure’s success level, and is it possible to classify it?	Proceed to Sect. 3 if the information for that specific section is explained
Findings	• Is there a concise description of the findings? • Would the results of the experiment be presented in the study?	Accept the research study if the results are appropriately mentioned; otherwise, reject it
Accuracy	• Is there some discussion on the primary accuracy metrics used? • Does the research make use of a predefined dataset for the analysis of comic book images?	If the questions in this section have been adequately addressed, accept the research study; if not, reject it

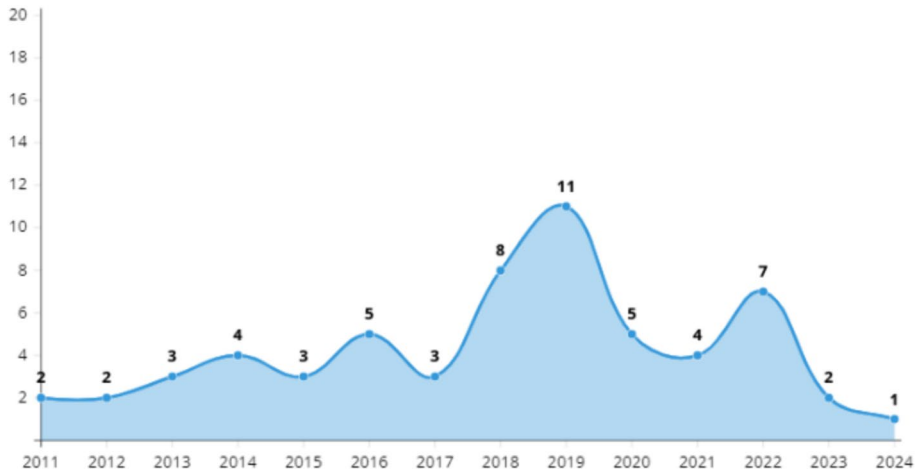


Fig. 7 Distribution of selected research studies

Table 7 Citation score

S. No	Citation	Count of Studies selected
1	0–20	30
2	21–40	16
3	41–60	11
4	61–80	0
5	81–100	2
6	101 and above	1

- **Narrative data synthesis method:** The term "narrative synthesis" refers to a technique [65] for systematically reviewing and synthesizing results from various studies that rely primarily on using text and words to describe the summarized results. The authors have examined the data and presented their findings as charts and tables. RQ1, RQ2, RQ3, RQ8, RQ9, and RQ10 are discussed this way.
- **Vote counting synthesis method:** Voting is a simple approach for combining data [66] from several analyses. It includes correlating the number of research studies that are positive with those that are negative. The authors have used this method for RQ4, RQ5, RQ6, and RQ7 to correlate two research studies. The authors in this scenario assessed the accuracy and determined which model outperformed the others.

3.6 Threats to validity

The authors have tried their best to find the appropriate studies using a search string phrased as per the research questions. Moreover, there may be a possibility that a few research studies have been missed by the authors who do not include the keywords in the abstract, title, and keywords in their research study. Although the authors made every effort

Table 8 Credibility score

Credibility Score(Range)	Selected Studies
1–2	S26, S28, S34, S46
3–4	S8, S9, S12, S15, S16, S20, S25, S53, S56
5–6	S1, S2, S6, S10, S13, S14, S17, S19, S21, S24, S29, S33, S35, S36, S38, S40, S42, S44, S49, S51, S57, S59
7–8	S3, S11, S18, S22, S23, S27, S30, S31, S32, S37, S39, S41, S43, S45, S47, S48, S50, S2, S54, S55, S58
9–10	S4, S5, S7, S60

to exclude this risk by consulting the bibliographies of individual research to identify all appropriate studies, there is still a possibility that we may have missed some relevant studies that may be regarded as a threat.

4 Results and discussion

The computational model offers a framework within which the information from prior studies can be extracted and summarized. The research studies are initially gathered from several search databases. After picking the research studies, the authors read each and collected relevant information. This information will help find various analysis techniques, parameters, journal details, and publication years for comic images. During the discussion, the authors analyzed the outcomes of findings within the framework of research questions along with a greater perspective that is more closely linked with the research questions. Some relevant research studies are also presented to support and justify the findings. The authors describe the findings for each research question derived from primary research studies in this section.

4.1 RQ1: What are the different parameters of comic images used to detect the information from comic images?

There are different parameters of comic images which can be used to fetch information from the comic images, such as speech balloons, gutters, text-boxes, panels, characters, and onomatopoeia [3]. Between these parameters, there is either no structural uniformity or very little. Therefore, it is challenging to segregate these features and get information from the images of comic book pages [55]. Each panel comprises speech balloons, characters, and text boxes with different shapes and forms. The story's text is often found in narrative text boxes and speech balloons. Speech balloons depict both characters' thoughts and dialogue between comic book characters. These speech balloons connect to their respective speakers via a tail pointer [75]. Therefore, the textual information in speech balloons facilitates users' understanding of the content of the comic story. Comic book images document mixed content handled differently based on the examined aspects. The techniques used might differ significantly depending on whether the emphasis is placed on balloons, panels, comic characters, or words. The authors review all these in the following sub-sections.

Table 9 Summary of Selected Studies (highly cited) on CR

Selected Studies	Summary	Advantages	Limitation
S15	The authors have proposed a method for extracting comic characters from the manga page collected from the Manga 109 dataset	- This proposed method automatically extracts and detects the Japanese comic characters from the page - Experimental results show that the authors have achieved 100% accuracy for flat comic panels and 93.75% for extracting Japanese comic characters	The study offers a commendable approach to extracting text from Japanese comics, but it does have some limitations. These include issues with text extraction accuracy, a narrow focus on a specific language and style of comics, questions about how well the system performs in real-time, and the need for additional refinement and expansion to make the system more applicable beyond its current scope
S25	The research emphasizes closure's function in linking panels and inferring unseen events in comics' distinctive narrative structure. The authors introduce the COMICS dataset, which contains over 1.2 million panels, and emphasize the need for computer models to understand text and image modalities to understand closure-driven narratives in stylized artwork and dialogue	- Examining COMICS shows that text or graphics cannot adequately convey a comic book story - Three cloze-style tasks are offered to provide a more thorough knowledge of the narrative structure, which tests the models' ability to anticipate narrative and character-centric characteristics of a panel given the context of previous panels	Several constraints have been highlighted, including the need for an enhanced understanding of context in computer models, the difficulties of integrating ordinary information, and the persistent difficulty of obtaining human-level performance in comics interpretation
S41	The authors have proposed a method based on mathematical morphology and region growing for automatically extracting the panels from comic pages and also offered a method for detecting speech balloons	The proposed method yields excellent outcomes, i.e. achieved 94.67% (Precision) and 75.53% (Recall)	- Much effort is necessary to improve the analysis of complex page types - The study used a minimal dataset for their experimental work

Table 10 No. of research questions answered

Reference	Selected Studies #Studies	Research Questions									
		RQ1	RQ2	RQ3	RQ4	RQ5	RQ6	RQ7	RQ8	RQ9	RQ10
[54]	S1	✓		✓	✓	✓					✓
[67]	S2	✓	✓	✓							✓
[28]	S3	✓		✓							✓
[3]	S4	✓	✓	✓	✓			✓	✓	✓	
[10]	S5	✓	✓		✓	✓	✓	✓		✓	✓
[68]	S6	✓			✓			✓		✓	
[9]	S7	✓	✓	✓	✓	✓	✓	✓	✓		✓
[69]	S8	✓	✓				✓	✓			
[47]	S9						✓	✓			
[14]	S10				✓	✓					✓
[13]	S11		✓	✓							
[51]	S12	✓	✓		✓				✓		
[70]	S13			✓	✓	✓					
[8]	S14	✓			✓	✓			✓		✓
[71]	S15	✓		✓	✓						✓
[31]	S16			✓	✓	✓					✓
[16]	S17	✓		✓		✓					✓
[53]	S18	✓			✓	✓					✓
[72]	S19			✓		✓					✓
[62]	S20			✓	✓	✓					✓
[17]	S21	✓		✓	✓						✓
[56]	S22		✓	✓				✓	✓		✓
[60]	S23				✓		✓		✓		✓
[45]	S24	✓		✓	✓	✓					
[73]	S25	✓			✓					✓	✓
[74]	S26	✓		✓	✓						
[75]	S27	✓		✓	✓	✓				✓	
[76]	S28	✓					✓			✓	✓
[77]	S29	✓		✓	✓						
[32]	S30	✓	✓	✓				✓			
[11]	S31	✓	✓	✓		✓					
[78]	S32	✓		✓	✓	✓			✓	✓	
[38]	S33	✓		✓	✓			✓	✓		
[79]	S34	✓		✓	✓			✓			
[4]	S35	✓		✓	✓	✓					
[80]	S36	✓	✓		✓			✓			
[15]	S37	✓	✓		✓						✓
[81]	S38	✓		✓	✓		✓	✓			
[33]	S39	✓		✓	✓						✓
[12]	S40	✓	✓			✓	✓		✓		
[6]	S41	✓			✓						✓
[82]	S42	✓			✓	✓		✓	✓		
[83]	S43	✓	✓		✓			✓	✓		

Table 10 (continued)

Reference	Selected Studies #Studies	Research Questions									
		RQ1	RQ2	RQ3	RQ4	RQ5	RQ6	RQ7	RQ8	RQ9	RQ10
[84]	S44	✓	✓	✓	✓						
[85]	S45		✓	✓	✓						
[55]	S46		✓	✓	✓						✓
[86]	S47		✓	✓					✓		
[21]	S48	✓		✓	✓						✓
[87]	S49	✓			✓	✓		✓			
[88]	S50	✓		✓	✓						✓
[89]	S51	✓	✓					✓	✓		
[90]	S52					✓					
[62]	S53	✓		✓			✓				✓
[91]	S54	✓		✓	✓						
[61]	S55	✓		✓				✓			
[57]	S56	✓	✓	✓				✓	✓		
[92]	S57	✓		✓		✓		✓	✓		
[93]	S58			✓		✓					
[94]	S59				✓			✓	✓		
[95]	S60	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

4.1.1 Panels

A comic book page consists of multiple panels used to define certain situations or actions. The panels may be divided by a gutter-styled white gap and encased in a frame [32]. Depending on the language, the panels are read in a particular sequence. For example, reading is often done from right to left and top to bottom in Japanese. Panels in the comic book contain captions and speech balloons, which describe the content of the comic story [96]. The panels' reading and dialogue balloons' reading orders are often identical. The authors have used panel detection to divide the comic page into different frames or panels. A panel is a picture section depicting a single scenario [35]. The bounding box will be specified as close to the frame as feasible, and they can be framed. An example of panels on an Indian comic image is shown in Fig. 8.

4.1.2 Speech balloon

Speech balloons are essential to the layout of comic pictures because they serve as visual constraints for the spoken emotions of characters [55], allowing conversation and spoken interactions to be conveyed, as shown in Fig. 9. Usually circular and with visible tails, these balloons are placed close to the character's lips to identify clearly which speaker is speaking simultaneously and to maintain consistency in situations with numerous conversations. For the best readability, font style consistency is essential [56]. Bold, italicized, or styled text may highlight specific phrases or communicate subtle tones. Careful attention is paid to the font choice, which complements the comic's overall tone and enhances reading and visual coherence [54]. Speech balloon colour schemes, which are often single and

intended to stand out against the backdrop, may be consistent for individual characters or can change to reflect changes in their emotional tones. Tailless balloons may express inner thoughts; tailless captions can provide additional story details [57]. Careful consideration of the padding and whitespace within the balloons is essential for a clean appearance and improved readability. In other words, speech balloons are crucial in defining the visual and narrative aspects of comics and helping to articulate conversation.

4.1.3 Comic characters

An individual explicitly created to tell a story inside the comic art form is known as a comic character, as shown in Fig. 10. Skilled comic book artists meticulously develop these characters' physical characteristics, personalities, and narratives, which are then imaginatively communicated via a combination of visual and literary elements [97]. Marvel heroes with superpowers, such as ordinary people dealing with complex problems and even anthropomorphic creatures, make up many comic book characters. Their complexity of growth, distinctiveness of traits, and audience reliability often determine the resonance and appeal of these characters [72]. Famous superheroes like Spider-Man, Batman, and Wonder Woman are good examples; they all have their backstory, problems with ethics, and special abilities. Carefully crafted comic characters not only add to a comic series' overall appeal, but they also reach out to readers on a deeper level, becoming cultural icons long after the comics have ended [11].

4.1.4 Narrative text boxes

Narrative text boxes are used to narrate [98]. They are additionally known as captions that include narrative material. However, both narrative text boxes and speech balloons' bounding boxes for textual information are referred to as text boxes, as shown in Fig. 11. As a result, the text boxes may partially overlap the speech balloon and narrative text box borders. Therefore, for a comic to be automatically understood, the information in the narrative text boxes and speech balloons is essential [54]. Text boxes can be used for various purposes, such as narration, descriptions, scene setup, or occasionally to express the thoughts or emotions of characters. This enables comics to move beyond being limited to character conversation [21]. The text box allows comic book creators to include details that wouldn't flow naturally into the story's dialogue.



Fig. 8 Comic Panel

4.1.5 Gutters

The gutter is the area that separates the panels, as shown in Fig. 12. If each panel represents a concept [73], it acts as a blank space between ideas, giving the reader time to process the first panel's information before linking it to the next. Gutter width is often constant, implying a seamless transition from panel to panel with no accentuation or change to the reading [88]. It might be a good idea to create space between scenes by giving the panel borders extra width, giving the viewer more time and room to relax and reflect on the plot.

4.1.6 Text

A significant part of the narrative, the dialogue, and the feeling of the story are conveyed by the text included in comic pictures, as shown in Fig. 13. This text is often found inside speech balloons, captions, and sound effects [99]. The words that characters say are enclosed inside speech balloons, and the tails of the speech balloons often point in the speaker's direction [87]. This helps readers associate the conversation with certain characters. Using different font styles, such as strong or italicized text, allows for the manifestation of subtle sentiments and highlights certain words or tones. A deeper level of narrative is achieved by using captions, which are shown outside the panels and provide more background, narration, or interior thoughts [81]. To express aural aspects like explosions and crashes, sound effects, represented by onomatopoeic words inside stylized writing, add to the dynamic graphic experience. Keeping a smooth flow of narrative and ensuring that readers can easily follow along and immerse themselves in the comic world requires the incorporation of text and artwork in a seamless way. The overall aesthetic and communication efficiency of the text in comic pictures is improved by carefully considering the text's font types, colours, and positioning [21].

When taken as a whole, the individual components that make up a comic image serve as invaluable channels via which the narrative's content may be comprehended and interpreted. The spatial arrangement of panels determines the chronological flow of the story, highlighting significant events and facilitating an organized reading experience. Panels also serve to emphasize crucial moments. Providing more contexts, speech balloons, thought balloons and captions contain the characters' conversation and provide insights into their

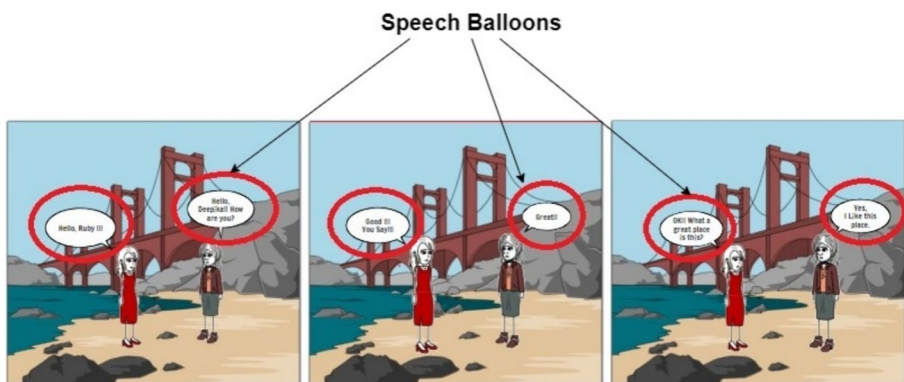


Fig. 9 Speech Balloon

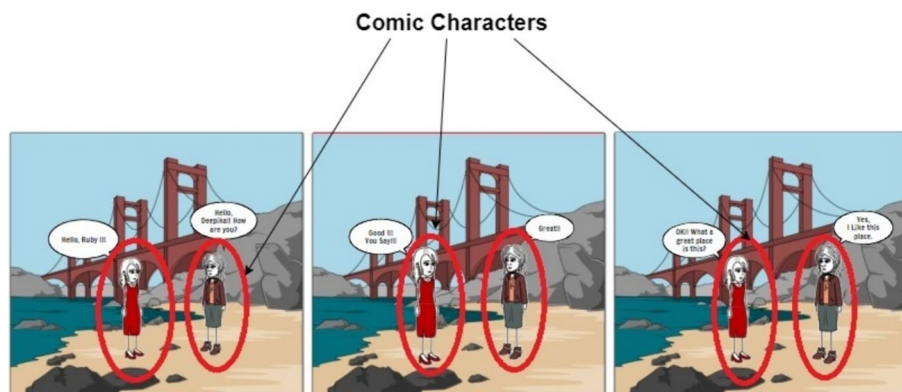


Fig. 10 Comic characters

interior thoughts. Additionally, these elements improve the story by providing additional information. Characters, which may be identified by their visual designs, facial expressions, and body language, play a crucial part in the story since they operate as channels through which roles are defined and emotions are conveyed. The parameters that have been stated play an essential part in the area of comic analysis and substantially influence the efficiency with which information is shared and the level of engagement that readers experience. Table 11 shows the in-depth analysis of all these parameters and how these parameters help to understand the content of the comic more effectively.

As discussed above, all these parameters play crucial roles in understanding comic content, so the authors have extracted various elements from comic images to extract relevant information. Researchers have removed no parameters from the comic images, as shown in Table 12. For example, in study S1, the authors utilized 8,911 speech balloons and text boxes from the comic image. Similarly, in the following study, S2, the authors extracted 453 speech balloons from the comic image; the same as in the other studies, the authors extracted the required elements to understand the content of the comic, which is given in Table 12.

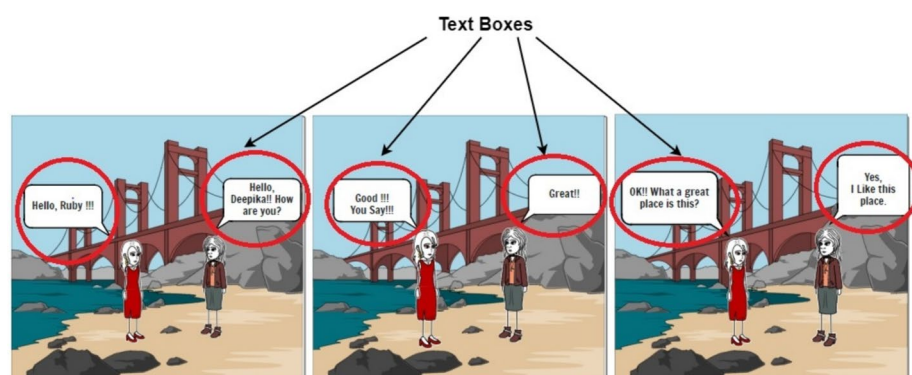


Fig. 11 Text boxes

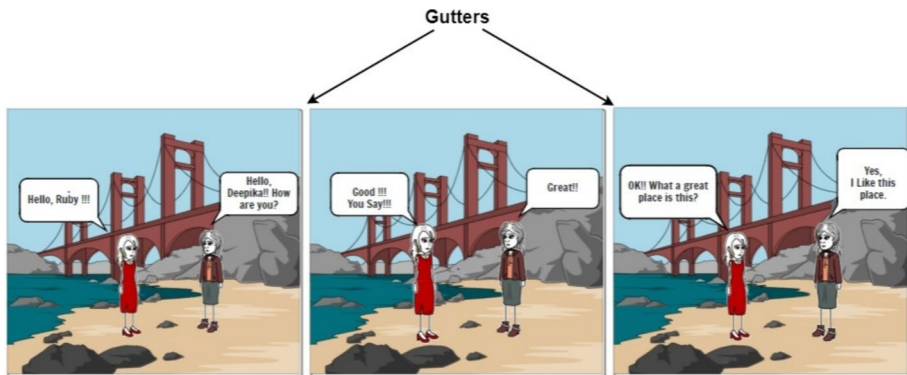


Fig. 12 Gutters (space between panels)

Challenges A variety of issues arise due to the absence of structural regularity among the elements that make up comic visuals. Because every parameter, such as speech balloons, gutters, text boxes, panels, characters, and onomatopoeia, contains distinct characteristics and forms, separating and extracting information from these many elements is complex. Because there is no standardized structure, there is either no consistency or very little uniformity among these components. Challenges that are associated with a lack of structural uniformity are:

- **Different layout:** The lack of structural consistency across comic features, such as speech balloons, gutters, and panels, leads to various layouts. This is because these elements are not uniformly structured. It is difficult for algorithms to recognize and comprehend the spatial links between these components due to the variety of the data, which affects the accuracy of the extracted information.
- **Variable Text Placement:** Within a panel, text may appear as speech bubbles, captions, and text boxes at various points. It is difficult to reliably extract textual information when there is no standard location, which impacts how well machines recognize and comprehend narrative aspects.

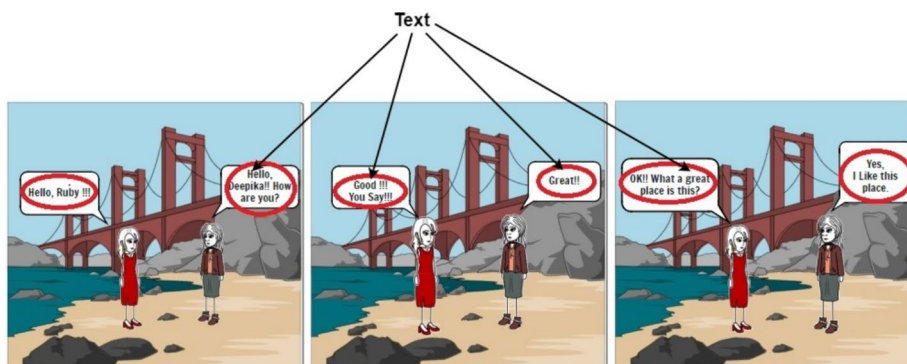


Fig. 13 Text inside speech balloons

- **Unconventional Panel Arrangement:** Panels might be organized in unusual ways, even if meant to define situations or activities. Providing a unified method for panel identification and interpretation is challenging since algorithms have to deal with differences in panel sizes, reading orders, and sequencing.

Summary of RQ1 The importance of each parameter in interpreting comic images is addressed. The issues linked with the absence of structural homogeneity among these aspects are among the most important discoveries. Additionally, the relevance of each parameter in transmitting narrative details, visual aesthetics, and overall storytelling dynamics is brought to light.

4.2 RQ2 What are the different types of speech balloons used in comic images?

Many distinct kinds of speech balloons may be seen in comic book illustrations. Each style of speech balloon is intended to represent a particular component of a character's words or thoughts [100]. The variety of available speech balloon kinds makes the representation of talks more visually exciting and nuanced. Listed below in Table 13 are numerous varieties of speech balloons that are often seen in comic book illustrations. The context of the conversation, the emotional tone being expressed, and the creative style of the comic all play a role in determining the sort of speech balloon that should be used. Comics are characterized by their visual storytelling, which is enhanced by the variety of speech balloons and adds to the overall narrative dynamics [82].

Summary of RQ2 All speech balloons express some contextual information and emotional arousal from the comic image. Among all the different types of speech balloons, the most used speech balloons in the comic image are rounded balloons, thought balloons and giant jagged contour balloons.

4.3 RQ3 What are the different computer vision techniques used for comic analysis?

The area of computer vision is a broad field that enables computers to acquire a high degree of understanding from digital pictures or movies. Its goal is to imitate the capacity of the human visual system to analyze visual input and to make judgments based on that interpretation for itself [81]. The primary goal is to make it possible for computers to acquire the ability to see, comprehend, and respond to visual input in a way comparable to how humans see [9]. The associated tasks in computer vision of semantic segmentation, object recognition, and instance segmentation are essential for the visual analysis of comics. Finding every object inside a group of object categories in an image and localizing it with its bounding boxes are necessary for object identification [48]. Each pixel in the picture is classified by semantic segmentation as belonging to one of many categories, often producing a significantly more precise delineation of object shapes. As an illustration, the result of semantic segmentation may be a collection of masks for panels, captions, and speech balloons [102]. Additionally, several instances of the same class are individually tagged in instance segmentation. Many segmentation techniques are used by the selected research studies, as shown in Table 14.

Table 11 Impact and effectiveness of the parameters

Parameter	Impact	Effectiveness
Panel	The flow of the story and the pace at which it unfolds are affected by the panels' layout. The chosen panel size, location, and sequencing influence the reader's perception of time and space inside the narrative	By directing attention, generating focus, and managing the pace of the narrative, panel layouts that have been well-researched may make storytelling more effective. Unorthodox layouts might be used to express a feeling of disorientation or to depict the characters' inner thoughts
Gutters	Understanding time and space between panels becomes much more difficult without gutters. A visual pause is provided, and readers can cognitively correlate several pictures	The speed of the tale is affected by the shape of the gutters and their placement. Wider gutters may signal a more extended time gap, while more undersized gutters suggest that transitions occur more quickly. It is possible to improve the entire reading experience by practising effective gutter management
Speech Balloon	In addition to contributing context and dimension to the images, text components communicate extra information, opinions, or conversation	Captioning and well-written speech balloons assist in the development of characters, the explanation of the storyline, and the story's emotional tone. It is essential to balance text and graphics when it comes to maintaining a dynamic and exciting reading experience
Text	Characters can communicate and express themselves mainly via dialogue (text). One's view of a character's personality, voice, and interactions with others is influenced by this parameter of the story	A character's growth cannot occur without a conversation that has been carefully designed. Establishing unique character voices, advancing the story, and communicating information in a natural and entertaining way are all accomplished via this
Comic characters	The reader develops a personal connection with the characters, resulting in the formation of emotional attachments that drive their involvement in the story	Readers can form a deep emotional connection with well-developed characters with characteristics, difficulties, and motives that are relevant to them. This sense of identification enhances the reader's ability to get more immersed in the story

Table 12 Parameters of the comic image

Selected Study #Studies	Semantic content in comics						Comic Page
	Speech Balloon	Narrative Text boxes	Comic Character	Gutters	Panels	Text lines	
S1	✓(8911)	✓					
S2	✓(453)						
S3					✓		
S4	✓(1092)		✓(1550)		✓(850)	✓(4691)	
S5	✓(1092)		✓(1550)		✓(850)	✓(4691)	
S6	✓(1092)	✓(1000)	✓(1000)		✓(1000)		
S7	✓(880)		✓(4.8 K)		✓(2 K)		
S8	✓	✓					
S9							✓(6,855)
S10			✓(1,550)		✓(850)		
S11	✓						
S12			✓(23,985)				
S13							
S14						✓(4691)	
S15	✓(70)						✓(712)
S16					✓(64)		
S17					✓(1,69,421)		
S18	✓(1,081)						
S19			✓(10,985)				
S20					✓(1,03,900)		
S21	✓(1,081)				✓(4500)		
S22	✓(2000)						
S23							✓(6,012)
S24	✓(1092)		✓(1,550)		✓(850)	✓(4691)	✓(194)
S25		✓(2,498,657)			✓(1,229,664)		✓(198,657)

Table 12 (continued)

Selected Study	Semantic content in comics						
	Speech Balloon	Narrative Text boxes	Comic Character	Gutters	Panels	Text lines	Comic Page
S26							✓(34)
S27	✓(1092)		✓(1,550)		✓(850)	✓(4691)	
S28							✓(5)
S29						✓(3537)	✓(800)
S30	✓(1092)		✓(1550)		✓(850)	✓(4691)	
S31			✓				
S32							✓(1300)
S33							✓(200)
S34			✓(115)				✓(1000)
S35			✓(91,924)				
S36					✓(121)		
S37					✓(371)		
S38							✓(126)
S39							✓(102)
S40					✓(1,03,900)		
S41					✓(355)		✓(42)
S42	✓(2025)						
S43	✓(372)						
S44							✓(1000)
S45	✓(1846)						
S46	✓(75)						
S47							
S48							✓(188)
S49		✓(2,208,006)				✓(15,345)	✓(20,000)

Table 12 (continued)

Selected Study	Semantic content in comics							
	#Studies	Speech Balloon	Narrative Text boxes	Comic Character	Gutters	Panels	Text lines	Comic Page
S50						✓(60)		✓(10)
S51								
S52		Created a tool for character design and creation of comic strip						
S53		32 Participants (Age between 10–11 years old)						
S54						✓(1,03,900)		
S55						✓(3740)		
S56		✓(1092)		✓(1550)				✓(1716)
S57								
S58								✓(5)
S59								✓(94)
		Developed digital comic media and their influence on improving communication skills						

Summary of RQ3 The author identifies different computer vision techniques used to analyse the comic image discussed above. Among all these techniques, the authors have used segmentation techniques (40%), ML techniques (28%), and DL techniques (32%).

4.4 RQ4 What are different non-computer vision techniques used for comic image analysis?

The term "non-computer vision techniques" refers to various methodologies used to analyse data and address issues without focusing on graphic information or image processing, as shown in Table 15. In contrast to computer vision, which primarily works with analyzing and extracting insights from visual data, non-computer vision approaches expand their applicability across numerous areas within Artificial Intelligence (AI), ML, and classical data analysis.

4.5 RQ5 What are different computational approaches to comics?

Although comics have been studied in the humanities for a while, computational methods of comic analysis are still in their early stages [46]. As a result, computational methods have hardly touched any of the categories that Cohn and Magliano's framework proposes. Most existing research focuses on visual and morphological structures, which are presentation-related elements at the unit level. Most sequential characteristics, notably story and event structure, are still widely ignored in computational comic analysis and higher-level

Table 13 Types of speech balloon

Reference No.	Name of Balloon	Type of Balloon	Definition
[54], [67], [59]	Thought balloon		The balloon's tail comprises a series of bubbles and resembles a fluffy cloud. The balloon is not circular; prominences in many directions may produce sub-balloons or linked sub-balloons; no bubbles-tail or more than one bubbles-tail; and the balloon is non-rounded.
[81], [53]	Rounded balloon		The balloon is an even, continuous-lined, smooth oval or circular. Some prominences may create sub-balloons or linked sub-balloons in one or more directions. There may also be no tail or even several tails.
[101], [83], [84]	Big-jagged contour balloon		Sharp-edged bulges on the balloon provide a regular or uneven shape. There may be no tail or more than one tail on such balloons, or the balloons could be joined or partially merged to make a complicated multi-balloon structure.
[83], [82], [56]	Serrated contour balloon		The balloon has an oval shape with a straight, finely sharp edge. The balloon is not spherical; protrusions exist in different directions, perhaps producing sub-balloons or linked balloons; there is no tail or more than one tail, and the balloon has non-rounded prominences.
[83], [84]	Interrupted contour balloon		The interrupted contour balloon is a smooth oval or circle with a dotted or irregular edge.
[83], [17]	Rectangular straight balloon		A square or rectangular balloon with rounded corners and straight edges.
[101]	Protruding edges Balloon		The balloon has four evenly spaced sharp projecting edges symmetrically arranged over a smooth oval or circle.
[83], [13]	Rectangular rounded balloon		The shape of a balloon is square or rectangular, with rounded corners and slightly protruding edges.

conceptual structure[71]. Computer vision is required for a computational study of comics to comprehend page layout, visual scenarios within and outside of individual panels, as well as people, objects, activities, and emotional expressions. To process story and dialogue, multimodal representations for text-image relationships, and high-level narrative models for recording events, narrative arcs, inter-character relationships, etc., natural language processing must also develop and maintain these models [45]. The three main categories of computational comic methodologies are content generation, content analysis, and user engagement, as shown in Fig. 14.

4.5.1 Content analysis

The content of comics must be extracted to be understood and used for services like retrieval or recommender systems. The DIA community started using conventional methods to address this issue. Screen tones[103] and text [71] are examples of low-level image analysis, whereas style [104] and genre [60] identification are examples of high-level analysis. Some features, like speech balloons and text, depend on one another since the former includes the latter. The placements may be related since the speech balloon often originates from a character's mouth. Usually, but not always, these components are arranged in a panel. The great variety among all comics makes the analysis challenging because authors are free to draw however they like. For instance, some authors exaggerate the facial distortion of a character's face to make him appear more angry or astonished [3].

4.5.2 Content generation

The purpose of content generation or enrichment is to use comic books to create new material based on comic books or other media [45]. Comics may be utilized as an input or an output to create new material in the following ways: comics to comics, comics to other media, and other media to comics [3]. The layout and placement of the various elements, such as the characters, text balloons, etc., to give a seamless reading experience are two challenges in creating comics.

4.5.3 User interaction

In addition to content analysis and creation, the authors have also established a separate area of study based on user engagement with comics [3]. Instead of studying comics, one section examines the user. For instance, the author liked to be able to recognize or anticipate the user's emotions or actions when reading comics. Another aspect is making new interfaces or ways for readers and comics to interact. The accessibility for individuals with disabilities can also be improved through modern technologies[105].

Summary of RQ5 Because of the intricacy of some layouts and the variety of information, segmenting the panels and reading comics' text remain challenging. A thorough analysis of the user's interactions with the comics has not been conducted. Numerous sensors may be used to assess users, including pupil dilation, eye movement, heart rate, perspiration, muscle activity, body posture, and pupil activity. More knowledge about readers and comics may be gained by gathering such data.

Table 14 Computer vision techniques

Computer Vision Technique	Segmentation technique	Pros	Cons	Selected Studies
Segmentation	Edge Detection	- Facilitates the localization of an object's borders inside an image - Contributes to design and structural analysis	- It is noise-sensitive - The efficiency of edge detection might be dependent on the image resolution	S2, S4, S45
	Region Growing	- Conceptually, it's easy to understand and use - It can deal with the low-level noise of images	- May experience either over-segmentation or under-segmentation - The homogeneity criteria are essential to the success of the region's growth	S5, S41, S44, S46
	Thresholding	- It is computationally simple and efficient - It may work well in different lighting environments - Many distinct types of images might benefit from thresholding	- Images with multimodal intensity distributions could be challenging for thresholding - Conventional thresholding techniques can only handle images with one channel	S7, S13, S14, S15, S16, S27, S34, S35
	YOLO	- Well-known for its ability to analyze data in real-time - It is flexible and may be used for many object-detecting tasks	- It may find it difficult to identify objects with a fine-grained - Potential difficulties with objects that overlap one another	S3, S17, S18, S32
Grab Cut		It works effectively when there is a strong contrast between the displayed object and the background	It may not be easy to distinguish when objects and backgrounds have similar colours or textures	S19, S24, S31, S48
		It is beneficial for foreground extraction jobs	For the best results, manual engagement is necessary	
	Optical character recognition (OCR)	- It helps to convert the text, either printed or scanned, to readable form - It allows to extract the structured data	It may have accuracy concerns, particularly with complicated layouts and handwritten text	S26, S29
Maximally Stable Extremal Region (MSER)		- It does not need to be manually initialized - It is quite simple to implement - In addition to having a high repeatability, MSER areas are easily distinguishable	- Performance is highly dependent on the parameters that are chosen - It may need a significant amount of computational power	S33, S36, S37, S39

Table 14 (continued)

Computer Vision Technique	Segmentation technique	Pros	Cons	Selected Studies
Machine Learning (ML)	SVM	• It possesses the ability to manage non-linear decision boundaries • Shows good performance with datasets ranging from small to medium in size	• It may be costly regarding memory, especially when dealing with large datasets • It is mainly intended for use in binary classification	S6, S8, S28, S38, S42
	Naïve byes	• The implementation of it is easy • It is effective for dealing with large datasets	• It may have difficulties when dealing with discrete or continuous data	S43
	Decision Tree	• It is simple to learn and interpret • It is capable of managing data that is both numerical and categorical	• Data is vulnerable to little fluctuations • They are not suitable for online learning configurations	S8
	Random Forest	• It often has a high degree of predictive accuracy • can effectively manage outliers and missing values • It is capable of collecting data that is both numerical and categorical	• There is less interpretability with these models • It has to be stored for numerous trees • These models are more complex to comprehend	S8

Table 14 (continued)

Computer Vision Technique	Segmentation technique	Pros	Cons	Selected Studies
Deep Learning (DL)	CNN	• CNNs can automatically learn hierarchical representations of features based on their given data • Resilient to variations in the supplied data	• Training deep convolutional neural networks (CNNs) might require significant computational effort • CNN, particularly when they have many parameters, is susceptible to overfitting	S1, S9, S10, S11, S23, S25, S40, S47
	R-CNN	• The R-CNN architecture is designed with modularity, enabling seamless integration of diverse components • CNN and its variations have exceptional proficiency in precisely localizing objects inside an image, offering accurate predictions for bounding boxes	• The initial version of the R-CNN model exhibits high computational complexity and suffers from slower training and inference durations	S12, S18, S20, S21, S32, S50, S30
	ACGAN	• The model can provide samples of superior quality and increased authenticity • It can produce various patterns of samples within a specific category	• R-CNN employs a multi-stage process that generates region proposals, extracts features, and classifies objects • The performance of a system may be influenced by hyperparameters, which are factors that determine the system's behaviour	S22, S49

Table 15 Non-CV techniques

Technique	Summary	Pros	Cons	Studies
Natural Language Processing (NLP)	The process of extracting and analyzing text from inside comics, including speech bubbles and captions	• Offers the reader a better understanding of the story framework and the conversation • Facilitates sentiment analysis to identify emotional tones	• Concerns about text that is either handwritten or styled • Having trouble with complicated layouts or speech balloon shapes that are not usual	S30, S54, S60
Graph Theory	Modelling the connections and interactions between comic book components (characters, objects, and situations)	• Captures visual narrative and connection inside the comic • It makes it possible to analyze thematic and structural relationships	• Need to define edges and nodes, which may be meaningless • The larger and more intricate the comic, the higher the level of complexity	S12, S45, S49, S60
Content-Based Image Retrieval (CBIR)	Indexing and obtaining images according to their visual content	• It makes it easier to get objects that are visually comparable efficiently • It helps examine recurrent themes and patterns in images	• Retrieval effectiveness depends on precise image annotations • Complex visual cues that are semantically significant may not be captured	S14, S59
Clustering algorithms	Assembling panels or photos that are similar to classify those using visual characteristics	• It helps to organize comics and provide insights into visual content • Allows for the classification of panels according to their visual similarity	• Continuity in the selection of clustering features • Potentially fails to convey narrative context or high-level semantics	S30, S45, S57
Interactive visualization	Exploring and presenting comic content in an interactive manner	• Exploration and user engagement with comic content are enhanced • Facilitates interactive navigation across scenes and panels	• Compelling visualizations need significant development and design work • User training may be required to optimize effectiveness	S15, S53
Human computation and crowdsourcing	Using crowdsourcing platforms or human annotators for subjective parts of comic analysis	• encapsulates the depth of human perception, comedy, and the emotional effect of the scenario • Enhances the ability to generate various annotations from a variety of angles	• Proper governance is necessary to guarantee the quality of annotations • Individual annotators may be responsible for subjectivity and prejudices	S54, S60

Table 15 (continued)

Technique	Summary	Pros	Cons	Studies
Data Mining	Using machine learning to analyze speech balloons, captions, and textual components for purposes such as sentiment analysis, topic modeling, and other similar purposes	• Learns from the text-based material that is included inside comics • It makes sentiment analysis easier to grasp to understand emotional tone better	• The level of accuracy may vary based on the level of complexity of the language that is employed • Attempting to generalize to various comic book genres can bring difficulties	S52, S55

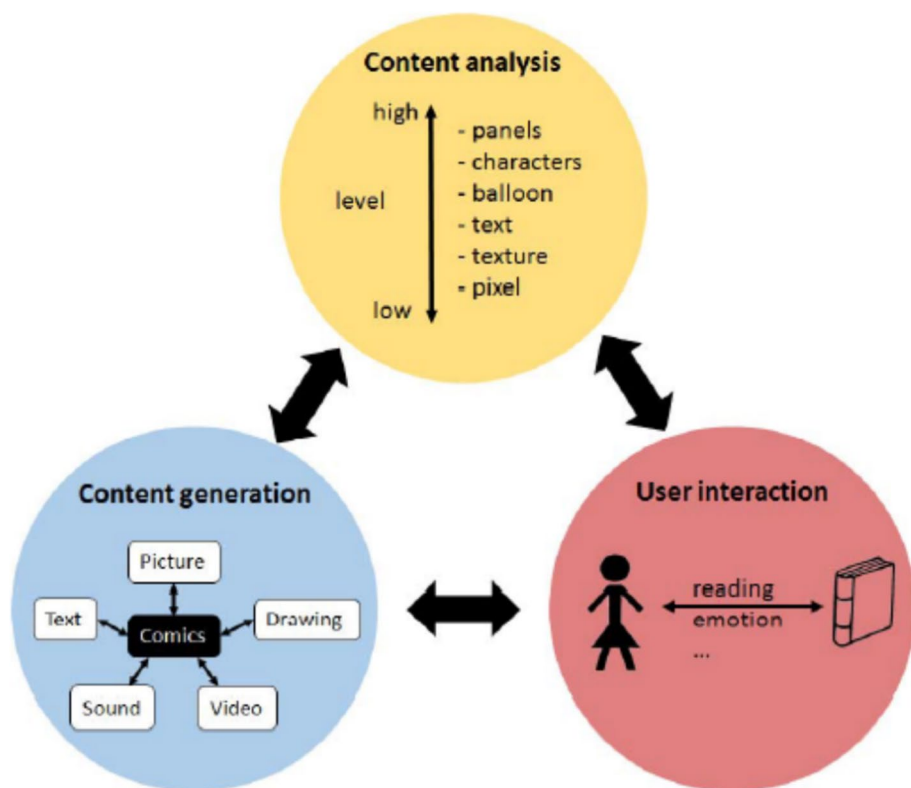


Fig. 14 Computational approaches [9]

4.6 RQ6 What are the different types of datasets available?

Due to copyright concerns, few datasets have been made freely available. However, no researcher is allowed to share and use copyrighted materials. It is, therefore, challenging to organize reproducible research and plan competition. Hopefully, several datasets have recently become available publically[12].

A few datasets are available for the researchers to use, which are eBDtheque, manga109, COMICS, BAM!, GNC, VLRC, and DCM_772, as shown in Table 16 with their detailed descriptions of how many comic books and pages a dataset contains [53]. Creating that kind of dataset requires much effort and money, especially when labelled and ground truth data are involved.

The main issue while creating these kinds of datasets is copyright and legal protection, due to which researchers are not authorized to make the datasets that will be available publically [3]. Depending on how the study develops, the dataset's content is also crucial. For instance, it's intriguing to have a selection of comics from many nations written in various genres and languages. The detailed description of each dataset is as follows:-

Table 16 Public Datasets

Dataset	Website	Publishing year	Title	No. of books	No. of pages	Images available publicly	Subtitle	Transcription of text
eBDtheque	http://ebdtheque.univ-lr.fr/registration/	1900–2010	American, French, and Japanese comics	25	100	Yes	Balloon, shape(polygon), panels, balloons, and text lines	Yes
COMICS	https://obj.umiacs.umd.edu/comics/index	1930–1950	American Golden Age comics	3,948	198,657	Yes*	Text boxes*, panels*	Yes*
Manga109	http://www.manga109.org/index_en.php	1970–2010	Japanese manga	109	21,142	Yes	Characters, text boxes, panels, and faces	Yes
DCM	https://gitlab.univ-lr.fr/crigau02/dcm-dataset/-/tree/master	1930–1950	American Golden Age comics	772	772	Yes	Faces, characters, and panels	Partly*
GNC	https://noveledebacle.humspice.ucla.edu/the-dataset/	1970–2010	Graphic novels in English	270	55,000	No	Captions, balloons, diegetic text, panels, and onomatopoeia	Yes
VLRC	https://figshare.com/articles/dataset/Data_statistics/910809/1	1940–2010	Comics from different genres and places	292	6,200	No	Page layout, panels, the semantic correlation between panels	No
BAM!	https://bam-dataset.org	-	Object labels and emotion labels	-	2.5 million	Yes	Various labels	Yes
BCBId	https://sites.google.com/view/banglacomibookdataset	2018 till date	Bangle comics	8	3,327	No	Speech balloons, panels, text, and characters	Yes

4.6.1 eBDtheque

eBDtheque was the first dataset that was made publicly available to the researchers. It was developed by L3Igroup at La Rochelle [35]. This dataset was published between 1905 and 2012. It consists of comic images written in Japanese, French, and American languages. A labelled element includes 1,550 characters, 4,691 test lines, 1,092 balloons, and 850 panels[32]. However, the number of images is limited. Also, developing such a type of labelled data is time-consuming, but it is very beneficial for society[54]. The dataset also consists of manual annotations of reading order, balloon shape, metadata on album artist, title, release date, etc., text lines, panel locations, and text transcription.

4.6.2 Manga109

This is another comic image dataset developed by the Aizawa Yamasaki Laboratory at the University of Tokyo [106]. It consists of 109 manga volumes that include various genres with 21,142 images. A manga artist wrote these in Japan in the source decade of the 1970s and 2010s. To make this dataset publicly available to the researchers, permission has been obtained from 94 professional creators [107]. Manga109 is a valuable choice for machine learning training material since it excels in annotation and size quality. The data set is only available for Manga, a genre with a distinct visual language and collection of aesthetic components[14]. As a result, training on Manga109 will not produce results that are easily transferable to comics from Europe or the US. A small amount of labelled data, such as the text for a few volumes, is available. This dataset's strength is that it provides all pages from a single volume, allowing for the analysis of page sequences [54].

4.6.3 Comics

This is another dataset of comics created by [73] in 2017. This dataset was created with the 4000 downloaded comic books from DCM, which had the highest rating. They moderately divided each page into panels and retrieved textbox positions from them [14]. Then, they used a commercial OCR service on the 2.5 million pictures of textboxes that resulted in transcribing the text. It comprises 1,229,664 panels from 3948 American comic books released between 1938 and 1954 and automated textbox transcriptions[98]. The dataset consists of 1500 textboxes and rectangular bounding boxes for panels on 500 pages of labelled data, among other ground truths. The annotated dataset can be downloaded from <https://github.com/miyyer/comics>. Although the amount of the data set is considerable, the quality of the data is limited because the majority of the segmentation was carried out by machines[1].

4.6.4 DCM

DCM stands for Digital Comic Museum. It is a digital collection of Golden Age comic books in the public domain released before 1959[54]. People thoroughly annotated another DCM subset. Rigaud, Burie, and Nguyen [10] developed a subset DCM_772, which contains 772 comic pages from DCM organized by publisher. This dataset may be appropriate for differentiated character recognition and detection because ground truth is present in the form of object-bounding boxes around characters[17], which are further classified into four classes: object-like, human-like, animal-like, and extra (supporting role characters).

4.6.5 VLRC

VLRC is Visual Language Research Corpus, another comic dataset created by Neil Cohn[1]. This dataset can be downloaded for research purposes from <http://www.visual-language-lab.com/vlrc/>. It contains 35,000 coded panels from almost 300 comics gathered from various locations (e.g. Hong Kong, Korea, The Netherlands, France, America, Sweden, Japan, Germany, and Belgium) between different time intervals (1940 –present). Also, it includes different genres[60]. It incorporates multimodality, exterior compositional structure (page layout), panel framing coding, semantic linkages between panels, and additional visual language structures. Human coders annotated each comic for at least 25 pages or 150 panels, with two coders annotating around 60% of the comics. The VLRC, utilized in various research on multimodality, page layout, narrative structure, and semantic structure[37], has the benefit of considering cultural variety across visual languages.

4.6.6 GNC

To establish a dataset helpful for humanities study, the Graphic Narrative Corpus [107], which aims to represent and strive to make a balance between prestige and popularity, was specifically chosen. The GNC, which concentrates on graphic novels in English, presently has over 270 volumes with about 55,000 pages. Most of the publications were released in Canada and the US, although graphic novels from other countries are also in the corpus [108]. Each work tells a continuous tale, is at least 64 pages long, and is intended for adults. The GNC includes works from the 1970s to the present that are fictional and non-fictional. Speech balloons, Polygons surrounding panels, captions, primary characters, diegetic text, and transcriptions, as well as onomatopoeia of all text objects, are all examples of human annotations on particular pages of each volume [78]. Translation, adaptations, Bibliographic details, awards given to specific works, and character cast lists are all included in the metadata. Eye movement records of at least ten readers of the first pages of each book are one of the GNC's distinctive features [7]. The free M3 annotation software was created by the same research team that created the GNC and can be found at https://groups.uni-paderborn.de/graphic-literature/wp/?page_id=3592.

4.6.7 BAM!

BAM stands for Behance Artistic Media Dataset. It comprises 2.5 million artistic images, including oil paintings, pen ink, comics, 3D computer graphics, watercolours, vector art, and pencil sketches. The dataset [109] comprises object labels that include people, dogs, cats, birds, buildings, bicycles, cars, trees, and flowers, and emotion labels that include scary, peaceful, gloomy, and happy. Various languages and content make this dataset more interesting[17].

4.6.8 BCBIId

For a while, comic books and graphic novels have drawn significant critical attention [54]. While there are several works on the "English comic genres," none have given attention to the "Bengali comic genres." A collection of 3327 images of various "Bengali Comic Books" from a wide range of prominent authors is included in this database [14]. This

database also includes the comic book of Narayan Debnath Samagro, 509 pages, which comprises all of the comics created by prominent author Narayan Debnath, including "Handa Bhodar Kamdokarkhana", "Nonte-Fonte", "Batul The Great" and among others. Similarly, Frank Cho has written a comic "Bonkonnya" which is 95 pages; another comic 'Feluda' written by Satyajit Ray (9 comics), and many more comics written by different authors are included in this database [62]. This dataset can be accessed only for research purposes, not for commercial use.

The authors have discussed all the datasets that are available for the researchers. Still, the authors have analyzed that in a few research studies, the researchers have used two or more datasets to check or enhance the accuracy of their model or technique [51]. So, in Table 17, the authors show the different datasets used by selected studies. In some research studies, the researchers have used some private datasets in which they have collected the comic images from the datasets that are discussed above or images collected from the internet. So, some private datasets used by the researchers in their research study are listed in Table 18 with the dataset's source.

Summary of RQ6 The most used popular public datasets are eBDtheque (55%), followed by manga109 (45%), and the most used private dataset is Sequencity612 (25%) and Fahad18 (20%), followed by comic images collected from internet or other source (55%).

4.7 RQ7 What are the different genres of comic images?

Worldwide, comics are a prevalent kind of entertainment. The notion of writers employing both text and illustrations is communicated through this medium. Even in a lifetime, it would be impossible to read all the comics published today [60]. Comic books are divided into genres to portray the features of the narrative; examples of these genres include fantasy, romance, and adventure. Using a single genre for an entire volume, however, makes it difficult to convey how the story differs from those of other comics in that genre [107]. Hence, even while modern comics search algorithms may locate well-known comics based on a genre for each volume, confident readers who would be fascinated by them will not be able to find such comics [110]. It is typical to think of a comic's story as scenes where specific things happen. One continuous scenario is drawn on one page, and then scenes vary on subsequent pages, which is how comic book artists frequently structure a comic book story. The researchers suggest [47] using the term "narrative structure" to refer to the arrangement of genres that make up a comic book story. Studying these comic stories is now possible using several standard methods (machine learning, classification algorithms, etc.).

There are 12 different types of genres defined in the Manga 109 dataset, as shown in Table 19, where this dataset is available for academic use. In Manga109, "BakuretsuKungFugirl" is an example of narrative structure [3]. The romantic comedy has been designated as the genre throughout "BakuretsuKungFugirl." Each page may fall under a different genre, such as humour, conflict, love and romance, and so on. Narratives are created by the order of the pages, which feature a variety of genres, as shown in Fig. 15.

Love Romance, Humor, Fantasy, Science Fiction, Battle, Sports, Romantic Comedy, Historical Drama, Animal, Suspense, Four Panel Cartoon, and Horror are among the 12 genres [111, 112]. The term "Four Panels Cartoon" describes a type of animation where each episode is wrapped up in four panels, regardless of how the panels are arranged.

Table 17 Comic images/pages from public datasets

Selected Studies		Mostly used Datasets (Public datasets)							
#Study	eBDtheque	Manga109	BAM!	VLRC	GNC	DCM	COMICS	BCBId	DCM_772
S1	100 pages	21,142 images				772 images		1373 images of 39 stories	
S2	50 pages								
S4	100 pages	21,146 pages	2.5 million artistic images				500 pages, 1500 textboxes		
S5	100 pages					3188 raw comic images			772 images
S6			21.1 K images						
S7	100 pages	21,142 images			6200 pages	55,000 pages	500 pages, 1500 textboxes		772 images
S8		21,142 images							
S9		6,855 pages							
S10	100 pages	21,142 images					500 pages, 1500 textboxes	980 images	
S11					750 annotated pages				
S12	100 pages								
S13	100 pages								
S14	100 pages								
S15		15 pages							
S16	Eight pages								
S18	100 pages	10,130 pages					1,98,657 pages		
S20		10,130 pages							

Table 17 (continued)

Selected Studies Mostly used Datasets (Public datasets)									
#Study	eBDtheque	Manga109	BAM!	VLRC	GNC	DCM	BCBId	DCM_772	
S21	100 pages							772 images	
S24	100 pages	194 images							
S25									
S26	100 pages								
S27	100 pages								
S28	Live participants (5) user's emotions were recorded with an eye tracker								
S29	46 pages								
S30	100 pages								
S31									
S39		104 pages							
S40		10,130 pages							
S41	100 pages								
S44	93 pages								
S48		20,000 pages							
S53		21,142 images							
S54						772 images			
S56	100 pages								
S57		5 pages							
S58	94 pages								

Table 18 Private Datasets

Selected Studies		Private datasets	
#Study	Name of dataset	Source	No. of pages/images taken
S3	Dragon Ball, vol. 42	-	237 images
S5	Japanese and Hong Kong comic books	-	2000 images
S6	Fahad18	obtained from Google Images by the authors [48]	586 images of 18 favourite cartoon characters
S12	Synth_90K	generated from a set of 90 k common English words	9 million images
S14	JC2463	Manga titles from Manga 109	14 manga titles
S17	AEC912	collected from the Internet	American and European comic pages
	Sequenciy612	https://www.sequenciy.com	612 pages
	Dragon ball	-	237 comic images
	Comic3w	Collected from the Manga 109 dataset	29,485 images
	Comic2k	collected from 40 comic books, which come from ten different Japanese and Hong Kong comics	2000 images
S19	Naruto, Vols. 46- 50, Dragon Ball (kanzenban) Vols. 29–34, Boys Over Flowers, Vols. 16–20, and Space Brothers, Vols. 1–5	-	10,985 images
S20	Microsoft coco	https://cocodataset.org/	over 330,000 images
S29	Sequenciy	https://www.sequenciy.com/	400 French albums
S31	Comic Images	Collected from China's digital comics	-
S32	Sequenciy 612	https://www.sequenciy.com/	612 images
	Fahad18	obtained from Google Images by the authors [48]	586 images of 18 favourite cartoon characters
	Ho42	-	42 comic pages
	Sun60	Taken from the eBDtheque dataset	60 comic pages
S33	Comic Images	Images collected from the internet	200 images
S34	Comic Images	Images collected from the internet	115 images

Table 18 (continued)

Selected Studies		Private datasets	
#Study	Name of dataset	Source	No. of pages/images taken
S35	Comic Images	100 comic images from each comic series published in different countries	1000 images
S36	Comic Images	Collected from web-comics ZenPencils	Four comic stories
S37	Comic Images	Web-comic set	94 pages
S38	Comic Images	Collected from several issues of distinct series	126 pages
S41	Comic Images	Extracted from 7 comic books	42 pages
S42	Comic Images	Images collected from the comic book “One Piece.”	2025 images
S44	Comic Images	Obtained from Japanese comic book	900 images
S45	Speech balloon	Collected from different publishers with several artistic styles	12 comic books
S46	Speech balloon	User evaluation experiment	Three male participants
S47	Comic Images	Collected from the Japanese comic book “Compeito! 1”	188 comic pages
S49	Comic text +	Images collected from the COMIC dataset	22,864 images
S50	User evaluation experiment (four pupils and a teacher in Grade 3)		
S51	Created a tool for character design and creation of comic strip		
S52	32 Participants (Age between 10–11 years old)		
S55	1716 different comic books collected from the internet		
S57	User evaluation experiment (collected bio signal from 11 subjects using EMOTIVPRO software)		
S59	32 students from English class for developing the digital comic media		

Table 19 Genres included in Manga 109

Different genres included in Manga 109

Love Romance	Battle	Humour	Romantic Comedy	Suspense	Historical drama
Animal	Horror	Suspense	Sports	4 Panels cartoon	Science fiction



Fig. 15 Genres images collected from the Manga 109 dataset. The genres depicted in the image are referenced from comic *BakuretsuKungFuGirl* by Ueda Miki, featured in the Manga109s dataset. Courtesy of Ueda Miki. Please see <https://www.manga109.org/> for more details

Summary of RQ7 The various genres of comic images have been discussed. Even the characteristics of other genres may be combined in certain comics to produce innovative and exciting storylines.

4.8 RQ8 How and which emotion of a character can be recognized through comic images?

Exciting information can be fetched from comics. There are many types of comics, including romance, comedy, horror, and others, and they all have various emotional effects [81]. Numerous studies have been conducted on identifying emotions when engrossed in videos using physiological signals like the electroencephalogram (EEG) and facial images [45]. Such studies have not, however, been carried out when reading comic books. Typically, emotions are shown as two axes: arousal and valence, where arousal refers to an emotion's intensity and valence to its positive or negative nature. The author demonstrated that it is feasible to gauge a reader's arousal by examining their physiological signals. In another research study, the author's emotion [76] demonstrated that it is feasible to infer whether a user is reading a romance, a comedy, or a horror comic based on the emotions felt by the readers by capturing and analyzing the physiological signals of the readers [3]. For instance, the user experiences stress when reading a horror comic book, and his skin temperature drops while his electrodermal activity (sweating) increases. There are two different ways through which the emotion of the comic character can be recognized, as shown in Fig. 16.

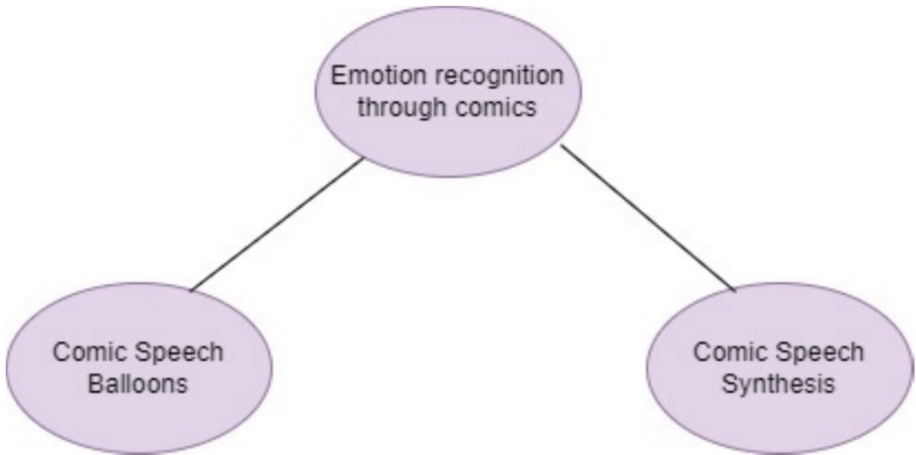


Fig. 16 Ways to detect emotions

4.8.1 Comic speech balloons

The speech balloon is an essential element of the comic, conveying most of the conversation between the leading characters. The balloon’s shape also reveals the speaker’s emotions [56]. Consequently, identifying the balloons and connecting them to the speaker is crucial. Narrative text boxes and speech balloons often include information about the story. Speech balloons depict character thoughts and conversations between characters from comic books [84]. Each speech balloon contains a tail pointer that helps to link them to their appropriate speaker. Therefore, the textual information contained in speech balloons facilitates viewers in following the narrative and understanding the comic content [38].

Speech balloons have different shapes [79], and the shape of these speech balloons helps to detect the speaker’s emotion, as shown in Fig. 17. For example, if the shape of the

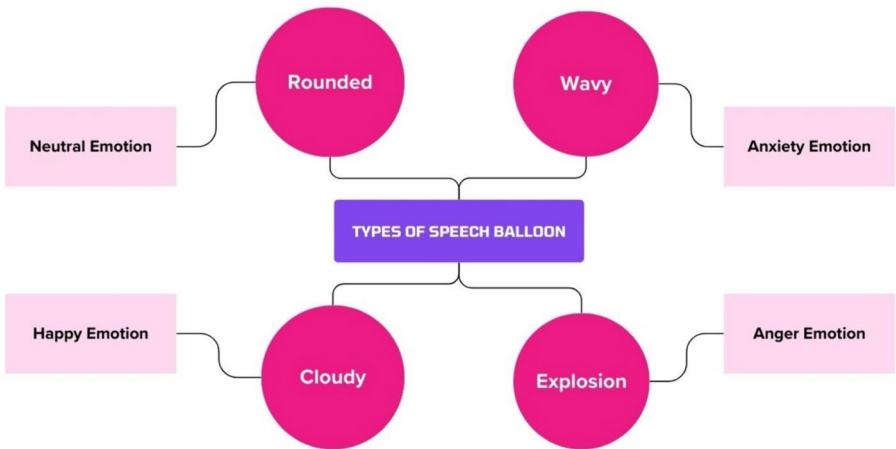


Fig. 17 Emotion associated with speech balloon

speech balloon is round, then it represents neutral emotion, and the speech balloon has a cloudy shape, which means it represents a positive emotional state, i.e. Joy or happiness; similarly, if the speech balloon has a wavy shape then it shows anxiety emotion, and if a speech balloon has an explosion shape then it shows the emotional state is anger [3].

4.8.2 Comic speech synthesis

A present scenario is producing audio comics similar to audiobooks, which have become more popular to increase the delight of comics. Current methods frequently use professional narrators who read comic books page by page to generate audio comics. In the research study [12], a professional narrator with the appropriate voice is hired, depending on the comic characters to be narrated. Suppose the comics feature multiple characters with varying ages, gender, and personality traits. In that case, several professional narrators may be hired to work together, each portraying one or more characters [74]. Throughout the narrating process, the narrators must pay great attention to the emotions of the comedic "speaker" by considering the facial expressions and sentences related to the character. The narrators must exert control over the tone and passion of their statements to produce a compelling, high-quality narration. To simplify and automate the production of audio comics, we suggest a computational approach. The strategy followed in the research study [68] is shown in Fig. 18. The author's methodology automatically narrates the characters with voices congruent with the visual content when given a comic page with various characters and words. This is accomplished through visual analysis of the comic.

Summary of RQ8 Nowadays, professional narrators who must pay close attention to the characters' emotions and manage their tone are used to create audio comics, which are growing in popularity. A computational solution that uses visual analysis to automatically narrate characters with congruent voices to streamline and automate production is described.

4.9 RQ9 Which Journal/Conference publishes the most comic detection and extraction research?

For several reasons, it is essential to look into the journals and conferences that regularly present research on comic recognition and extraction. It first assists in locating significant places where researchers provide noteworthy discoveries, making a thorough and focused

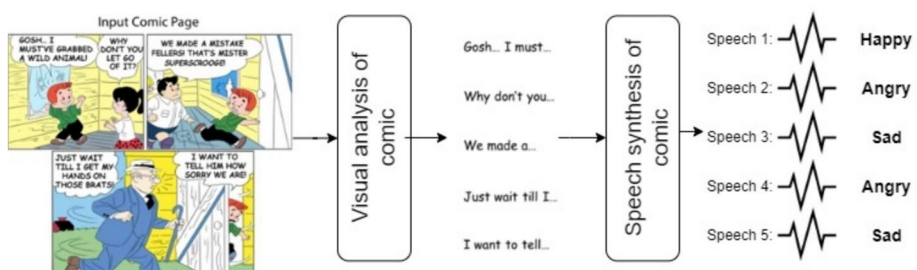


Fig. 18 Emotion detection through speech synthesis

literature search easier. Researcher attention may be directed on obtaining the most important studies and foundational works in the subject by identifying the primary options for such types of research. Incorporating reputable and significant sources into the literature review is further ensured by knowing the quality and importance of publications in reputed journals and conferences. This information also offers insights into new approaches and changing patterns within the CR research community. Knowing the main areas of interest also encourages networking and possible partnerships with subject matter experts, which adds a variety of viewpoints and levels of competence to the systematic review. A complete, up-to-date, and knowledgeable literature evaluation requires an exploration of the field of scientific publications in CR. Various dominant journals/ conferences are publishing research work on comic detection and extraction; Table 20 shows the list of conferences and journals publishing research on the topic “comic detection and extraction”. As observed from the table, 55% of the research studies publish their research work in the journal, while the rest of the 45% submit their research work at the conference.

Summary of RQ9 The most dominant journals publishing research in Comic Analysis are Multimedia Tools and Applications, Pattern Recognition, Journal of Imaging, International Journal on Document Analysis and Recognition, etc.

4.10 RQ10 What are the different accuracy measurements used for a comic analysis?

To conduct an SLR, it is essential to investigate the accuracy metrics used in CR. By providing insights into the criteria used to assess the effectiveness of CR algorithms, this study serves as a cornerstone for comprehending the methodological landscape of numerous investigations [62]. The researchers can undertake a comparison study by diving into precision, recall, and other metrics across various tasks, such as panel extraction and text recognition. This allows them to determine the advantages and disadvantages of multiple techniques [105]. This understanding not only makes it easier to identify crucial evaluation criteria but also makes it possible to conduct a sophisticated review of the quality of the study and to correlate the metrics selected with specific CR tasks. In addition, being aware of the accuracy metrics used is helpful in determining possible research gaps and areas in which standardized or more complete assessment procedures can be advantageous for the advancement of the discipline [39]. The authors of the selected research studies have used various measurement techniques to evaluate the efficiency of their developed model, which is listed below and in Table 21.

4.10.1 Precision

What proportions of cases are favourable out of all the projected positive instances? The model’s optimistic prediction from the entire dataset is used as the denominator in Eq. 1. Consider evaluating it to determine "to what extent the model is accurate when it claims it is right."

$$\text{Precision} = \frac{Tp}{Tp + Fp} \quad (1)$$

Table 20 Details of Journals/Conferences

Selected Studies	Name of Publication	Type
S13, S32	IAPR International Conference on Document Analysis and Recognition	Conference
S9, S26, S41	IAPR International Workshop on Document Analysis Systems	Conference
S25	IEEE Conference on Computer Vision and Pattern Recognition	Conference
S46	IEEE International Symposium on Mixed and Augmented Reality Adjunct	Conference
S20, S53	IEEE International Symposium on Multimedia (ISM)	Conference
S15	International Journal of Image Processing	Journal
S1, S21, S27	International Journal on Document Analysis and Recognition	Journal
S42	Interspeech 2014	Conference
S18	International Conference on Advances in ICT for Emerging Regions	Conference
S2, S11, S16, S23, S24, S31, S33, S35, S44	International Conference on Document Analysis and Recognition	Conference
S10, S14	International Conference on Document Analysis and Recognition Workshops	Conference
S38	International Conference on Signal-Image Technology & Internet-Based Systems	Conference
S34	International Conference on Pattern Recognition, Informatics, and Medical Engineering	Conference
S19	International Congress on Advanced Applied Informatics	Conference
S30	Journal of graphic novels and comics	Conference
S4,S5	Journal of imaging	Journal
S45	multimedia tools and applications	Journal
S3	Pattern Recognition	Journal
S50	Scandinavian Journal of Educational Research	Journal
S7	Topics in cognitive science	Journal
S6, S8, S17, S22, S28, S29, S36, S37, S39	ACM Digital	Journal
S47, S48, S49, S36	ACM Digital	Conference
S31	International Conference on Intelligent, Interactive Systems, and Applications	Conference
S40	Journal of Big Data	Journal
S51	Digital Creativity	Journal
S54	Procedia Computer Science	Journal
S55	IEEE Transactions on image processing	Journal
S56	International Conference on Computational Intelligence and Sustainable Engineering Solutions (CISES)	Conference
S57	IEICE Communications Express	Journal
S58	Lecture Notes in Computer Science	Conference
S59	Journal of Innovation in Educational and Cultural research	Journal
S60	Engineering Applications of Artificial Intelligence	Journal

Table 21 Evaluation criteria

Research Studies	Parameter	Dataset	Recall (%)	Precision (%)	F1 Score (%)	Accuracy (%)
S1	Speech Balloon	BCBId	97.07	98.80	97.92	
		eBDtheque	93.5	96.2	94.8	
	Narrative text boxes	DCM	92.7	96.9	94.7	
		Manga 109	93.56	95.49	94.51	
		DCM 772	94.50	97.22	95.84	
S2	Speech Balloon	BCBId	95.63	98.52	97.05	
		eBDtheque				
	Panel	Dragon Ball, vol. 42	98.58	98.38	98.78	
		Japanese and Hong Kong comics	95.82	97.64	96.73	
		Manga 109				92.5
S7	Speech Balloon					94.7
		Panel				
	Text characters and recognition		88.7	94.3	91.2	
		Character face detection		91.2		
						99.4
S8	Speech Balloon	Manga 109				92.06
		Comic page				
	Panel	eBDtheque	98	97	97.49	
		Manga 109	97.25	98.76	97.92	
		DCM	97.55	98.28	97.89	
S10	Comic Character	BCBId	98	98.55	98.24	
		eBDtheque	97	98.52	97.74	
	Speech balloon	Manga 109	98.71	99.14	98.82	
		DCM	98.23	98.71	98.41	
		BCBId	98.55	99	98.74	
S11	Speech balloon	GNC	94.04	95.58	94.48	

Table 21 (continued)

Research Studies	Parameter	Dataset	Recall (%)	Precision (%)	F1 Score (%)	Accuracy (%)
S12	Comic Character	eBDtheque	49.8	75.2	60.0	
		JC2463	93.2	95.0	94.1	
		AEC912	73.1	82.4	77.5	
S13	Text lines	eBDtheque				80.06
S14	Panel	eBDtheque	78.47	82.15	80.27	
		Sequency 612	82.04	86.94	84.42	
S15	Frame Extraction	Manga				96.88
	Speech balloon					98.86
	Text Extraction					93.75
S16	Comic character	eBDtheque				96.28
S17						
S18	Speech balloon	eBDtheque				0.89
S19	Comic characters	Naruto	0.625	0.833	0.714	
		Dragon Ball	0.166	0.666	0.265	
		Boys Over Flowers	0.583	0.368	0.451	
		Space Brothers	0.411	0.411	0.411	
S20	Panel	Manga 109	0.877	0.968		

Table 21 (continued)

Research Studies	Parameter	Dataset	Recall (%)	Precision (%)	F1 Score (%)	Accuracy (%)
S21	Speech balloon	DCM_772	89.50	94.87	92.11	87.06
		eBDtheque	74.94	92.77	82.91	
	Panel	DCM_772	98.71	96.84	97.76	
		eBDtheque	90.91	92.11	91.50	
	Comic Character	DCM_772	67.56	76.21	71.62	
		eBDtheque	62.17	71.79	66.63	
	Balloon character association	DCM_772	45.64	71.01	55.57	
		eBDtheque	28.40	71.93	40.35	
	Narrative text boxes	DCM_772	84.38	91.22	87.66	
		DCM_772	72.39	82.12	76.95	
S22	Character face	Manga 109	94.1	84.6	87.06	88.09
	Speech balloon	Manga 109				
	Comic page	COMICS				
	Comic character	eBDtheque				
	Comic character	eBDtheque				
	Panel	eBDtheque				
	Text localization	eBDtheque				
	Speech Balloon	eBDtheque				
	Comic page	user evaluation experiment				
	Text line localization	eBDtheque				
S28	Text recognition		75.8	76.2		

Table 21 (continued)

Research Studies	Parameter	Dataset	Recall (%)	Precision (%)	F1 Score (%)	Accuracy (%)
S30	Speech Balloon	eBDtheque	74.94	92.77	82.91	
	Panel		76.95	73.19	75.03	
	Comic character		62.17	71.79	66.63	
	Balloon-character association					
S31	Comic Character	Comic images	28.32	52.88	36.89	80.06
S32	Comic character	Sun60	35.48	79.43		92
		Ho42				
		Fahad_18		43.77		
S33	Text recognition	Comic Images				52
S34	Comic character	Comic Images				94.82
S35	Text localization	Comic Images	86.11	90.99	88.48	
S37	Comic Page	web-comic set				98.99
S38	Comic Page	Comic Images				91.26
S39	Comic page	Manga 109				91.3
S40	Panel	Manga 109			84	
S41	Comic page	Comic Images	75.53	94.67		
S42	Speech Balloon	Comic book				87.1
S43	Speech Balloon	eBDtheque				85.20
S44	Speech Balloon	eBDtheque	86.9	90.1	86.9	
S45	Speech Balloon	Comic Images				97.04
S47	Comic Images	Comic Images				0.62
S48	Comic page	Manga 109	75	83	76	
S49	Narrative text boxes	Comic text +	97.47	97.97		
S53	Panels	Manga 109	0.785	0.959		

Table 21 (continued)

Research Studies	Parameter	Dataset	Recall (%)	Precision (%)	F1 Score (%)	Accuracy (%)
S54	Panels	DCM	0.77	0.99	0.71	
S55	Panels	Comic images collected from the internet				
S56	Comic characters and Speech balloon	eBDtheque	93.10	91.83	92	
S57						80
S58	Text	eBDtheque	62.47	78.52		

4.10.2 Recall

Out of all genuine positive situations, what proportion of them is positive? The number of positive examples in the dataset is the denominator (TP + FN) in Eq. 2. Use it as an experiment to determine "how many more correct ones the model missed when it presented the right ones."

$$\text{Recall} = \frac{Tp}{Tp + Fn} \quad (2)$$

4.10.3 F1 Score

Percentage of unfavourable events relative to all unfortunate events. When there are negative cases in the collection, the denominator (TN + FP) equals their actual number. Although recollection is comparable, the emphasis is now on unfavourable events.

$$\text{F1score} = 2 * \frac{\text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}} \quad (3)$$

4.10.4 Accuracy

When a value is measured accurately, it is as near to the accurate measurement of the thing. In other words, the degree to which the estimated value is comparable to the reference or actual value. The accuracy may be assessed by gathering a large number of little readings. The low reading brings down the computation error.

$$\text{Accuracy} = \frac{\text{numberofcorrectprediction}}{\text{totalnumberofprediction}} \quad (4)$$

In the selected research studies, the authors used various classification and segmentation techniques to train their model and check its efficiency; they used accuracy measurements listed in Table 17. The table shows that the authors achieved good accuracy, precision, recall, and F1 Score in their research work.

Summary of RQ10 The authors have used different accuracy measures for evaluating the efficiency of their developed model or technique, such as precision, recall, and F1 score (54%), and they used accuracy measurement (46%).

5 Implications of CR

5.1 Theoretical implications

- **Story Interpretation:** CR improves the understanding of the narrative structures seen in comic sequences that enhance theory concepts. A comprehensive examination of narrative techniques and their effects on reader understanding is made possible by the automated extraction of panels, voice balloons, and text.

- **Semiotics and Graphic Representation:** Analyzing the visual components of comics broadens our understanding of semiotics and visual communication. Understanding the various components explains how layouts, colours, and symbols communicate meaning, providing insights into the relationship between visual language and communication theory.
- **Representations in Society and Culture:** Researchers may explore comics' social and cultural representations using CR. Comic stories provide an insight into cultural nuances, social viewpoints, and a variety of topics and this insight may be explored using automated visual content analysis.

5.2 Practical implications

- **Retrieving and Indexing Content:** CR allows for more efficient indexing and retrieval of content. With the use of automated panels and text recognition, searchable databases may be developed, making particular comic information more accessible and making it easier for researchers, adherents, and educators to retrieve.
- **Interactive Reading Tools:** CR has real-world applications in creating adaptive reading systems. It makes it possible to build intuitive interfaces that automatically detect comic image parameters that allow for the customization of presenting formats according to user preferences, thereby meeting the demands of people with varying reading habits and capabilities.
- **Educational Resources and Availability of Access:** Using CR as a teaching technique is very beneficial in educational institutions. It is possible to include it in educational platforms to automate comics analysis, providing professionals with insights into the materials used in learning. Additionally, it improves accessibility for those who have visual impairments by extracting text from the document.

6 Conclusion

With the growing popularity of digital comics, the dynamic field of CR has become a focus point with potentially revolutionary applications in the digital media industry. With a focus on 2011 through 2024, this systematic literature study explores the intricacies of CR techniques and approaches. The primary goal was to summarize important information on datasets, recognition methods, and the comparative context of recognition models. Upon analyzing 60 selected studies, a dichotomy regarding the application of recognition techniques became apparent. An extraordinary 65% of the studies preferred computer vision (CV) methods, demonstrating the prevalence and effectiveness of sophisticated visual algorithms. In addition, it is worth noting that 35% of the selected studies used non-CV techniques, highlighting the wide range of methodological approaches utilized in CR. For example, CV techniques that include Convolutional Neural Networks (CNNs) are very effective in learning hierarchical features, which helps them with tasks such as character recognition. Edge detection algorithms aid object outlining, while text recognition techniques, such as optical character recognition (OCR) and deep learning, are essential for extracting text from speech balloons. The use of region-based segmentation makes it easier to identify panels. In contrast, using object recognition models and deep learning for style analysis improves several elements of comic image analysis. Natural Language Processing and Content-Based Retrieval are examples of non-computer vision approaches used in comic analysis. These techniques

improve understanding by evaluating language, patterns, and user interactions. These techniques give significant insights that go beyond the conventional visual content. Through these methodologies, sentiment analysis, pattern identification, and user-centric approaches are enhanced, contributing to the comprehensive study of comic material. The analysis of the dataset landscape unveiled that 78% of the studies relied on established public datasets. Notably, the eBDtheque and Manga109 datasets emerged as the most frequently utilized. The fact that numerous studies have utilized this dataset demonstrates its status as a standard against which CR methodologies are evaluated and compared. Both Manga109 and eBDtheque are examples of comic datasets that are often used because of the enormous collections of manga and comic pages that they include. These collections provide researchers with a broad variety of visual styles, genres, and varieties of facts. The datasets include a wealth of annotations, making them useful for training and assessing algorithms in various tasks, including panel extraction, speech bubble identification, and text recognition. This is one of the reasons why they are so popular in the area of comic analysis research. Concurrently, a proportion of studies (22%) spearheaded the development of customized datasets, demonstrating the researchers' ability to adapt environments to the complex challenges of digital comic recognition. Despite these powerful techniques, navigating different layouts, changeable text locations, and unique panel configurations isn't easy when using CV approaches. However, the flexibility of algorithms is hindered by the intrinsic lack of structural homogeneity that exists throughout comic elements. However, non-CV approaches, despite their diversity, struggle with difficulties associated with their flexibility in accepting a variety of comedic structures. The challenges encountered by CV and non-CV approaches are exacerbated by the fact that comic layouts comprise a wide variety of creative styles. Innovative solutions are needed to address the difficulties mentioned above as the CR environment continues to change. To resolve structural anomalies in comic images, hybrid approaches, which aim to combine CV and non-CV procedures, provide an encouraging path forward. Comic layouts, text positioning, and panel patterns are all subject to change; thus, researchers should focus on creating adaptable algorithms to handle these challenges. The diversity of comic images presents significant challenges, yet hybrid techniques may overcome these issues and make comic identification systems more robust. Making a universal solution for comics is difficult because of the wide variety in creative style, layout, and text display. Because of the hybrid techniques, the system can easily adapt to different comic book layouts, graphic features, and text structures. When using a hybrid technique, adaptive learning techniques may be used. This allows the system to learn from new data and update its models constantly. This flexibility is essential because of the dynamic nature of comic book stories and visual styles. Through adaptive learning, the system can enhance its accuracy by learning from new instances and gaining an increased understanding of the unique characteristics and patterns seen in various comic books. The future of comic image analysis looks beyond the technical to include improved user involvement. Research directions that might take advantage of developments in human-computer interaction could focus on improving user experiences while engaging with digital comics. Another critical frontier is the examination of various comic image genres. A more sophisticated and genre-specific approach to CR may be possible if recognition algorithms can better adjust to the distinctive visual languages of other genres.

This in-depth investigation of CR offers an innovative strategy for further research and capturing the complexity of comic image analysis as it exists now. The difficulties that arise from structural anomalies stimulate creativity, especially in hybrid methods. As the digital comic medium develops, a future marked by sophistication, adaptability, and user-friendliness is anticipated through technological advancements, user-centric considerations, and a nuanced genre-specific approach.

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